Posttest 10

Multiple Choice

Identify the choice that best completes the statement or answers the question.

____ **1.** A branch of the _____ supplies the right atrium and right ventricle with blood.
- **a.** circumflex (Cx) artery
- **b.** right coronary artery (RCA)
- **c.** left main coronary artery
- **d.** left anterior descending (LAD) artery

____ **2.** Stimulation of parasympathetic nerve fibers typically results in which of the following actions?
- **a.** Constriction of coronary blood vessels
- **b.** Increased strength of cardiac muscle contraction
- **c.** Increased rate of discharge of the sinoatrial (SA) node
- **d.** Slowed conduction through the atrioventricular (AV) node

____ **3.** The contribution of blood that is added to the ventricles and results from atrial contraction is called
- **a.** afterload.
- **b.** atrial kick.
- **c.** cardiac output.
- **d.** peripheral resistance.

____ **4.** Which of the following are semilunar valves?
- **a.** Aortic and pulmonic
- **b.** Aortic and tricuspid
- **c.** Pulmonic and mitral
- **d.** Tricuspid and mitral

____ **5.** The left main coronary artery divides into the
- **a.** marginal and Cx branches.
- **b.** marginal and LAD branches.
- **c.** anterior and posterior descending branches.
- **d.** LAD and Cx branches.

____ **6.** _____ cells are specialized cells of the electrical conduction system responsible for the spontaneous generation and conduction of electrical impulses.
- **a.** Working
- **b.** Pacemaker
- **c.** Mechanical
- **d.** Contractile

____ **7.** The absolute refractory period
- **a.** begins with the onset of the P wave and terminates with the end of the QRS complex.
- **b.** begins with the onset of the QRS complex and terminates at approximately the apex of the T wave.
- **c.** begins with the onset of the QRS complex and terminates with the end of the T wave.
- **d.** begins with the onset of the P wave and terminates with the beginning of the QRS complex.

____ **8.** Which of the following statements is true regarding the QT interval?
- **a.** The QT interval represents atrial depolarization followed immediately by atrial systole.
- **b.** The QT interval corresponds to atrial depolarization and impulse delay in the AV node.
- **c.** The QT interval represents ventricular depolarization followed immediately by ventricular systole.
- **d.** The QT interval represents the time from initial depolarization of the ventricles to the end of ventricular repolarization.

____ **9.** How do you determine whether the atrial rhythm on an electrocardiogram (ECG) tracing is regular or irregular?
a. Compare QT intervals
b. Compare PR intervals
c. Compare R to R intervals
d. Compare P to P intervals

____ **10.** Which of the following ECG leads use two distinct electrodes, one of which is positive and the other negative?
a. Leads I, II, and III
b. Leads V_1, V_2, and V_3
c. Leads V_4, V_5, and V_6
d. Leads aVR, aVL, and aVF

____ **11.** In sinus arrhythmia, a gradual decreasing of the heart rate is usually associated with
a. expiration.
b. inspiration.
c. excessive caffeine intake.
d. early signs of heart failure.

____ **12.** An ECG rhythm strip shows a ventricular rate of 46 beats/min, a regular rhythm, a PR interval of 0.14 second, a QRS duration of 0.06 second, and one upright P wave before each QRS. This rhythm is
a. sinus arrest.
b. sinus rhythm.
c. SA block.
d. sinus bradycardia.

____ **13.** SA block is a disorder of impulse ______, and sinus arrest is a disorder of impulse ______.
a. formation, conduction
b. conduction, formation

____ **14.** Signs and symptoms experienced during a tachydysrhythmia are usually primarily related to
a. vasoconstriction.
b. atrial irritability.
c. slowed conduction through the AV node.
d. decreased ventricular filling time and stroke volume.

____ **15.** Which of the following correctly reflects examples of ectopic (latent) pacemakers?
a. The SA node and AV junction
b. The AV junction and ventricles
c. The SA node and right bundle branch
d. The AV junction and left bundle branch

____ **16.** A wandering atrial pacemaker rhythm with a ventricular rate of 60 to 100 beats/min may also be referred to as
a. atrial flutter.
b. atrial fibrillation (AFib).
c. multiform atrial rhythm.
d. multifocal atrial tachycardia.

____ **17.** The most common type of supraventricular tachycardia (SVT) is
a. atrial flutter.
b. atrial tachycardia.
c. AV reentrant tachycardia (AVRT).
d. AV nodal reentrant tachycardia (AVNRT).

____ **18.** Which of the following is true with regard to treatment of a symptomatic patient with AFib?
a. With rate control, the patient remains in AFib, but the ventricular rate is reduced (controlled) to decrease acute symptoms.
b. With rate control, measures are taken (either pharmacologic or electrical) to reestablish sinus rhythm.

____ **19.** If the onset or end of paroxysmal atrial tachycardia or paroxysmal supraventricular tachycardia (PSVT) is not observed on the ECG, the dysrhythmia is called
a. sinus tachycardia.
b. junctional tachycardia.
c. SVT.
d. multifocal atrial tachycardia.

____ **20.** Which of the following is the most common sustained dysrhythmia in adults?
a. AFib
b. Sinus bradycardia
c. Junctional rhythm
d. Ventricular tachycardia (VT)

____ **21.** Which of the following statements is true regarding the differences between premature atrial complexes (PACs) and premature junctional complexes (PJCs) in leads II, III, and aVF?
a. A PAC has a narrow QRS complex, and a PJC has a wide QRS complex.
b. A PAC has a negative P wave before the QRS complex, and a PJC has a positive P wave before each QRS complex.
c. A P wave may or may not be present with a PAC, whereas a PJC typically has a positive P wave before the QRS complex.
d. A PAC typically has a positive P wave before the QRS complex, whereas a P wave may or may not be present with a PJC.

____ **22.** A Wolff-Parkinson-White (WPW) pattern is associated with a
a. long PR interval, delta wave, and wide QRS complex.
b. short PR interval, delta wave, and wide QRS complex.
c. long PR interval, flutter waves, and narrow QRS complex.
d. short PR interval, flutter waves, and narrow QRS complex.

____ **23.** Which of the following ECG characteristics distinguishes atrial flutter from other atrial dysrhythmias?
a. The presence of fibrillatory waves
b. The presence of delta waves before the QRS
c. Clearly identifiable P waves of varying size and amplitude
d. The saw-tooth or picket-fence appearance of waveforms before the QRS

____ **24.** When a junctional rhythm is viewed in lead II, where is the location of the P wave on the ECG if atrial and ventricular depolarization occur simultaneously?
a. Before the QRS complex
b. Within the QRS complex
c. After the QRS complex

____ **25.** The usual rate of nonparoxysmal junctional tachycardia is
a. 50 to 80 beats/min.
b. 80 to 120 beats/min.
c. 101 to 140 beats/min.
d. 150 to 300 beats/min.

____ **26.** Junctional (or ventricular) complexes may come early (before the next expected sinus beat) or late (after the next expected sinus beat). If the complex is *early*, it is called a(n) _____. If the complex is *late*, it is called a(n) _____.
a. escape beat; premature complex
b. premature complex; escape beat

____ **27.** Depending on the severity of the patient's signs and symptoms, management of slow rhythms may require therapeutic intervention including
a. defibrillation.
b. administration of atropine.
c. synchronized cardioversion.
d. vagal maneuvers and administration of adenosine.

____ **28.** The term for three or more premature ventricular complexes (PVCs) occurring in a row at a rate of more than 100/min is
a. ventricular trigeminy.
b. ventricular fibrillation (VF).
c. a run of VT.
d. a run of ventricular escape beats.

____ **29.** PVCs that look alike in the same lead and begin from the same anatomic site (i.e., focus) are called _______ PVCs.
a. uniform
b. isolated
c. multiform
d. interpolated

____ **30.** Which of the following dysrhythmias has QRS complexes that vary in shape and amplitude from beat to beat and appear to twist from upright to negative or negative to upright and back, resembling a spindle?
a. AFib
b. Idioventricular rhythm
c. Polymorphic VT (PMVT)
d. Monomorphic VT

____ **31.** When a delay or interruption in impulse conduction from the atria to the ventricles occurs as a result of a transient or permanent anatomic or functional impairment, the resulting dysrhythmia is called a(n)
a. sinus arrest.
b. SA block.
c. bundle branch block (BBB).
d. AV block.

____ **32.** Whenever the criteria for BBB have been met and lead V_1 displays an rSR′ pattern, you should suspect a
a. left bundle branch block (LBBB).
b. right bundle branch block (RBBB).

____ **33.** The PR interval of a first-degree AV block
a. is constant and less than 0.12 second in duration.
b. is constant and more than 0.20 second in duration.
c. is generally progressive until a P wave appears without a QRS complex.
d. gradually decreases in duration until a P wave appears without a QRS complex.

____ **34.** Which of the following is an example of a complete AV block?
a. First-degree AV block
b. Second-degree AV block type I
c. Second-degree AV block type II
d. Third-degree AV block

____ **35.** Which of the following statements is correct regarding 2:1 AV block?
a. The atrial rhythm is irregular.
b. The PR interval remains constant.
c. The ventricular rate is twice the atrial rate.
d. The level of the block is located within the SA node.

____ **36.** The terms *advanced* or *high-grade* second-degree AV block may be used to describe one or more consecutive P waves that are not conducted.
a. True
b. False

____ **37.** Which lead is probably the best to use when differentiating between RBBB and LBBB?
a. V_1
b. V_4
c. II
d. aVR

____ **38.** The term *capture*, as it pertains to pacing, refers to
a. a vertical line on the ECG that indicates the pacemaker has discharged.
b. the extent to which an artificial pacemaker recognizes intrinsic cardiac electrical activity.
c. a pacemaker response in which the output pulse is suppressed when an intrinsic event is sensed.
d. the successful conduction of an artificial pacemaker's impulse through the myocardium, resulting in depolarization.

____ **39.** The 12-lead ECG only provides a _____-second view of each lead.
a. 1
b. 2.5
c. 4
d. 6

____ **40.** Although a right ventricular infarction may occur by itself, it is more commonly associated with a(n) _____ wall myocardial infarction (MI).
a. septal
b. lateral
c. inferior
d. anterior

____ **41.** *Poor R-wave progression* is a phrase used to describe R waves that decrease in size from V_1 to V_4. This is often seen in an _____ infarction.
a. inferobasal
b. anteroseptal
c. anterolateral
d. inferoposterior

Completion

Complete each statement.

42. The right atrium receives deoxygenated blood from the ____________________ ____________________ ____________________ (which carries blood from the head and upper extremities), the ______________ ____________________ ____________________ (which carries blood from the lower body), and the ____________________ ____________________ (which receives blood from the intracardiac circulation).

43. A beat originating from the AV junction that appears later than the next expected sinus beat is called a(n) ____________________ ____________________ ____________________.

44. A rapid, wide-QRS rhythm associated with pulselessness, shock, or heart failure should be presumed to be ____________________ ____________________.

45. PACs associated with a wide QRS complex are called ____________________ ____________________ PACs, indicating that conduction through the ventricles is abnormal.

46. ____________________ is the period of relaxation during which a heart chamber is filling.

47. The thick, muscular middle layer of the heart wall that contains the atrial and ventricular muscle fibers necessary for contraction is the ____________________.

48. Delivery of an electrical current timed for delivery during the QRS complex is called ____________________ ____________________.

49. Sometimes when a PAC occurs very prematurely and close to the T wave of the preceding beat, only a P wave may be seen with no QRS after it (appearing as a pause). This type of PAC is termed a(n) ____________________ PAC.

50. If the AV junction paces the heart, the electrical impulse must travel in a(n) ____________________ direction to activate the atria.

51. A demand pacemaker is also known as a(n) ________ pacemaker.

52. The axes of leads I, II, and III form an equilateral triangle with the heart at the center (Einthoven's triangle). If the augmented limb leads are added to this configuration and the axes of the six leads moved in a way in which they bisect each other, the result is the ____________________ ____________________ ____________________.

53. Indicate the heart surface viewed by each of the following.
Leads II, III, aVF: ____________________
Leads V_1, V_2: ____________________
Leads V_3, V_4: ____________________
Leads I, aVL, V_5, V_6: ____________________

Short Answer

54. Beginning with the right atrium, describe (by numbering) blood flow through the normal heart and lungs to the systemic circulation.

_____ Right atrium	_____ Left ventricle
_____ Mitral valve	_____ Pulmonary arteries
_____ Aorta	_____ Tricuspid valve
_____ Right ventricle	_____ Pulmonary veins
_____ Pulmonic valve	_____ Aortic valve
_____ Left atrium	_____ Systemic circulation

55. List five steps used in ECG rhythm analysis.
1. ____
2. ____
3. ____
4. ____
5. ____

56. Explain the benefits of a dual-chamber pacemaker.

____

____

____

57. List three uses for ECG monitoring.

1. ____

2. ____

3. ____

58. Describe the appearance of a pathologic Q wave.

____

____

____

59. What is the most important difference between sinus rhythm and sinus tachycardia?

____

____

____

60. List three causes of artifact on an ECG tracing.

1. ____

2. ____

3. ____

61. How do coarse and fine VF differ?

____

____

____

62. List four reasons why the AV junction may assume responsibility for pacing the heart.

1. ____

2. ____

3. ____

4. ____

63. Explain why patients who experience AFib are at increased risk of having a stroke.

____

____

____

64. On the ECG, what do the ST segment and T wave represent?

____

____

____

65. Fill in the blank areas in the table below.

ECG Finding	AVNRT	Atrial Flutter	Atrial Fibrillation
Rhythm			
Rate (beats/min)			
P waves (lead II)			
PR interval			
QRS duration			

66. Explain the difference between electrical capture and mechanical capture.

____

____

____

67. List four primary characteristics of cardiac cells.

1. ____

2. ____

3. ____

4. ____

68. Explain what is meant by the phrase "anatomically contiguous leads."

____

____

____

69. Indicate the ECG criteria for the following dysrhythmias.

	Second-Degree AV Block Type II	2:1 AV Block
Ventricular rhythm		
PR interval		
QRS width		

70. Fill in the blank areas in the table below.

ECG Finding	Idioventricular Rhythm	Accelerated Idioventricular Rhythm	Monomorphic Ventricular Tachycardia
Rhythm			
Rate (beats/min)			
P waves (lead II)			
PR interval			
QRS duration			

Posttest Rhythm Strips

Use the five steps of rhythm interpretation to interpret each of the following rhythm strips. All rhythms were recorded in lead II unless otherwise noted.

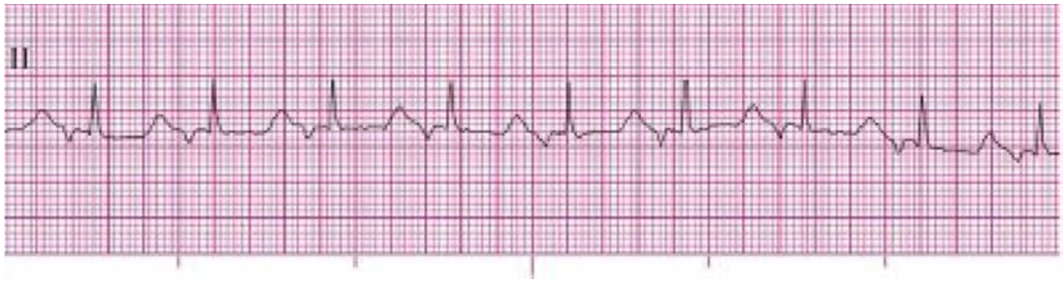

Fig. 10.1

71. **Fig. 10.1** This rhythm strip is from a 62-year-old woman with renal failure.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

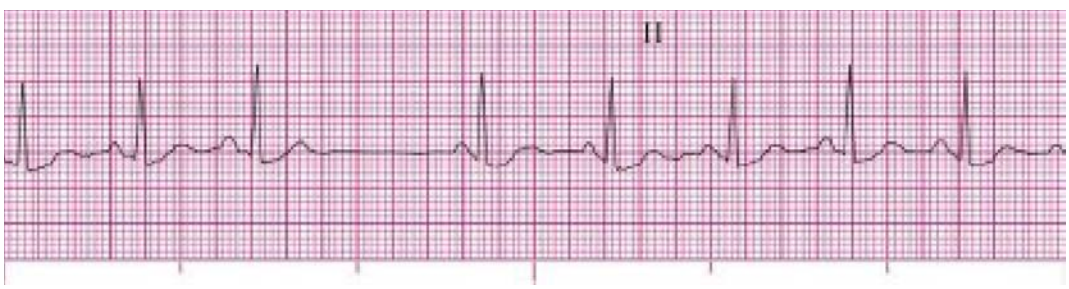

Fig. 10.2 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

72. **Fig. 10.2**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.3 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

73. **Fig. 10.3**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

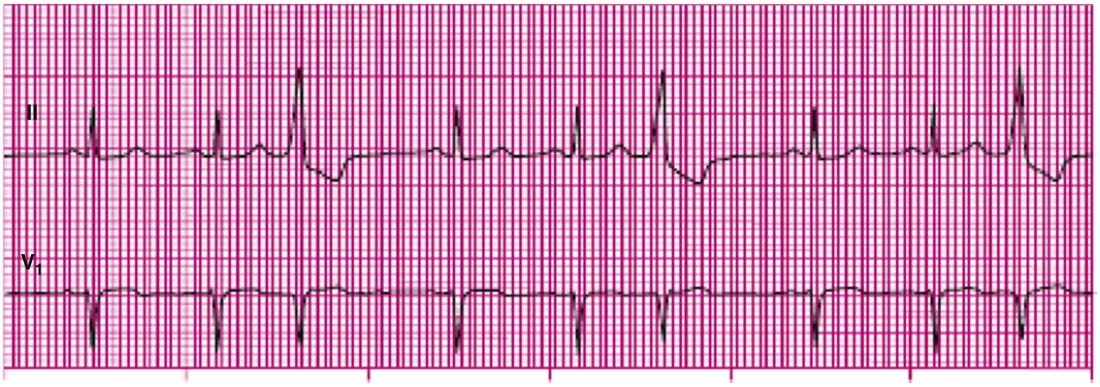

Fig. 10.4

74. **Fig. 10.4**

Rhythm: __________________ Rate: __________________ P waves: ___________________________________

PR interval: _______________ QRS duration: __________ QT interval: ________________________________

Interpretation: ____

Fig. 10.5

75. **Fig. 10.5** This rhythm strip is from a 59-year-old man complaining of poor circulation in his legs.

Rhythm: __________________ Rate: __________________ P waves: ___________________________________

PR interval: _______________ QRS duration: __________ QT interval: ________________________________

Interpretation: ____

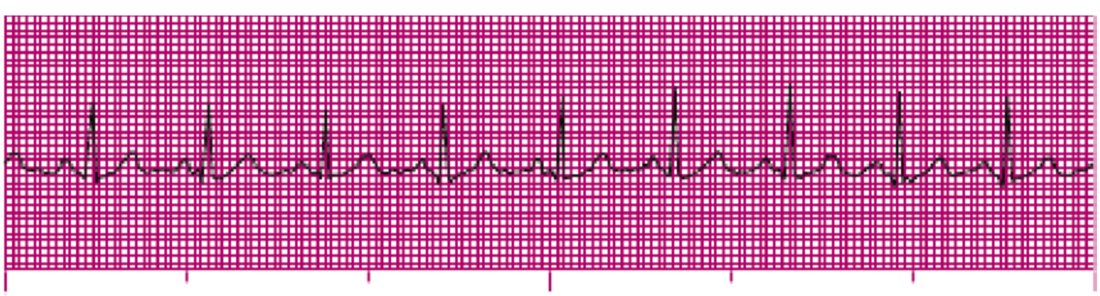

Fig. 10.6 (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

76. **Fig. 10.6** This rhythm strip is from a 32-year-old woman complaining of dizziness and shortness of breath.

Rhythm: __________________ Rate: __________________ P waves: ___________________________________

PR interval: _______________ QRS duration: __________ QT interval: ________________________________

Interpretation: ____

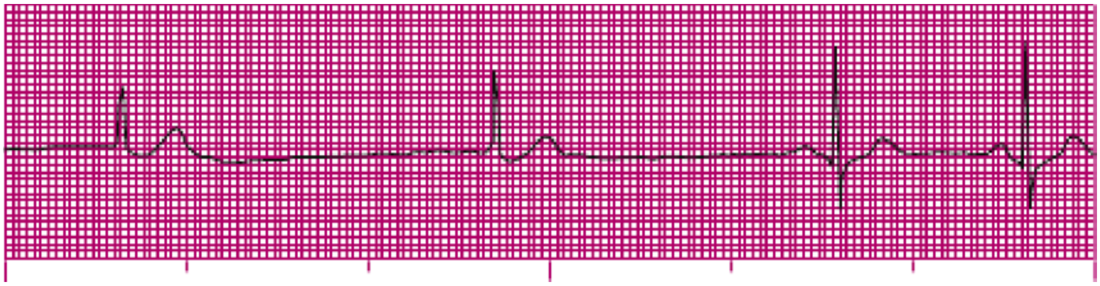

Fig. 10.7 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

77. **Fig. 10.7**

Rhythm: ________________ Rate: ________________ P waves: ________________________________

PR interval: ______________ QRS duration: _________ QT interval: ____________________________

Interpretation: ____

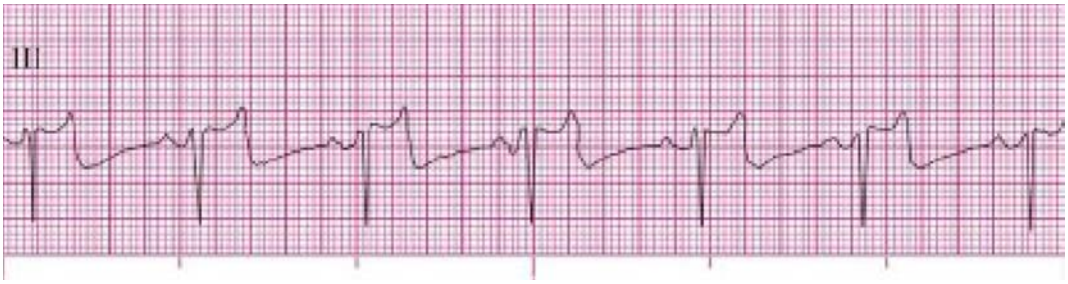

Fig. 10.8 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

78. **Fig. 10.8** This rhythm strip is from a 76-year-old man who experienced a syncopal episode while playing golf.

Rhythm: ________________ Rate: ________________ P waves: ________________________________

PR interval: ______________ QRS duration: _________ QT interval: ____________________________

Interpretation: ____

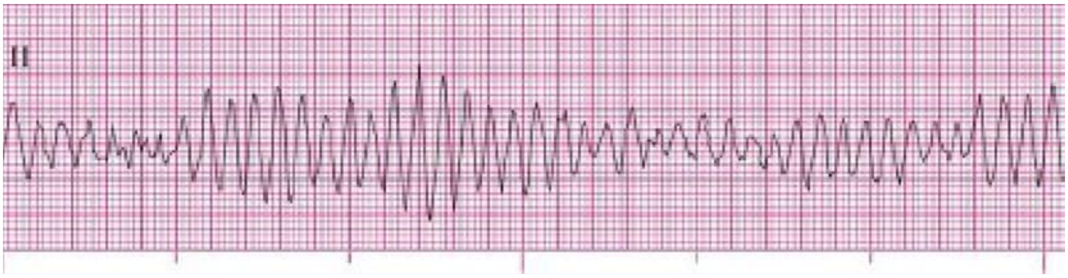

Fig. 10.9

79. **Fig. 10.9** This rhythm strip is from a 54-year-old man complaining of chest pain.

Rhythm: ________________ Rate: ________________ P waves: ________________________________

PR interval: ______________ QRS duration: _________ QT interval: ____________________________

Interpretation: ____

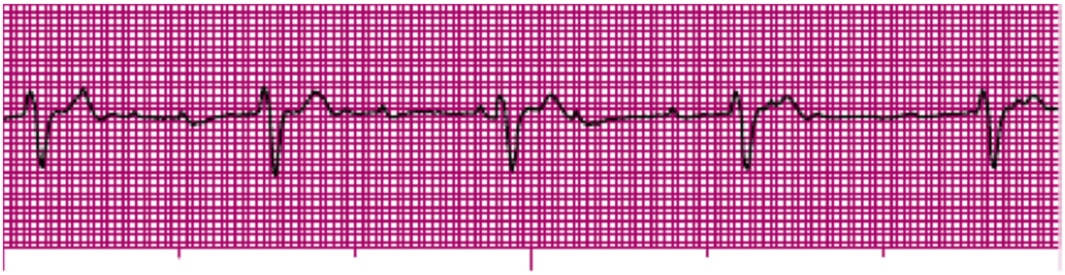

Fig. 10.10 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

80. **Fig. 10.10**

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

Fig. 10.11 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

81. **Fig. 10.11** This rhythm strip is from a 70-year-old woman with chronic obstructive pulmonary disease.

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

Fig. 10.12

82. **Fig. 10.12** This rhythm strip is from a 77-year-old man with chest pain. His chest hit the steering wheel during a motor vehicle crash.

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

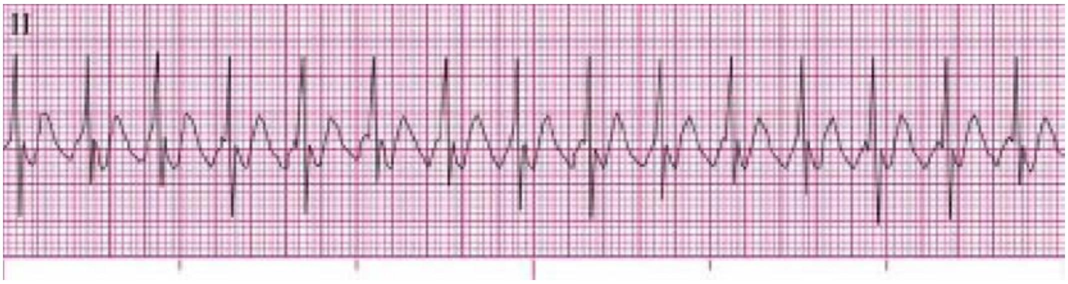

Fig. 10.13 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

83. **Fig. 10.13**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

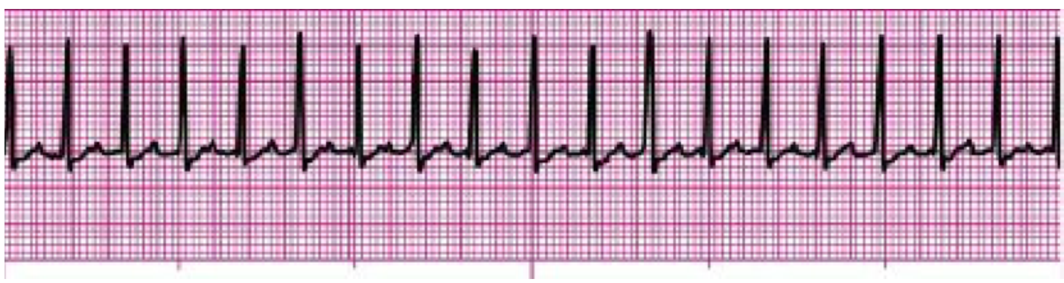

Fig. 10.14 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

84. **Fig. 10.14** This rhythm strip is from a 20-year-old woman who collapsed on the sidewalk of her residence. A family member states that she has a history of SVT and takes atenolol.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

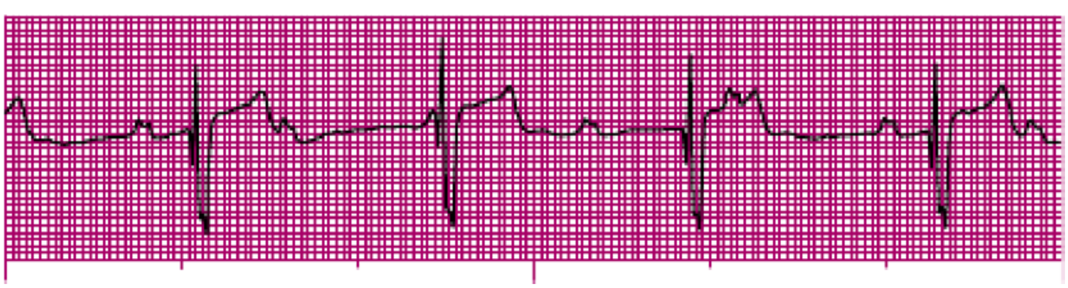

Fig. 10.15 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

85. **Fig. 10.15** This rhythm strip is from a 42-year-old man with chest pain.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

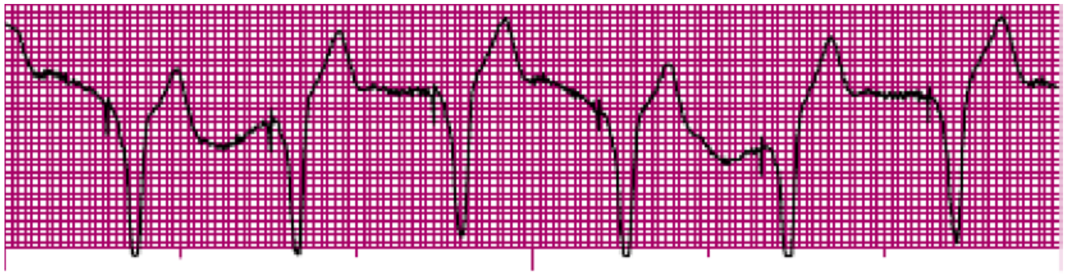

Fig. 10.16 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

86. **Fig. 10.16** This rhythm strip is from a 76-year-old woman who is complaining of back pain. Her medical history includes an MI 2 years ago.

Atrial paced activity? _______________ Ventricular paced activity? _____________________________________

Pacemaker malfunction? ___________ Identification ____

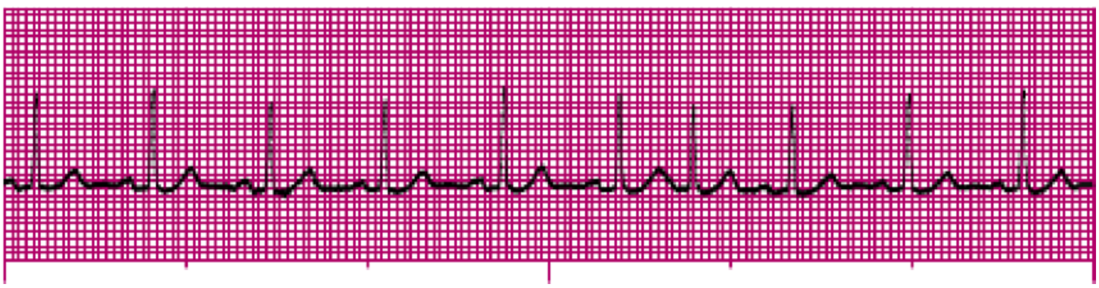

Fig. 10.17

87. **Fig. 10.17**

Rhythm: ___________________ Rate: ___________________ P waves: ____________________________________

PR interval: ________________ QRS duration: __________ QT interval: _________________________________

Interpretation: ____

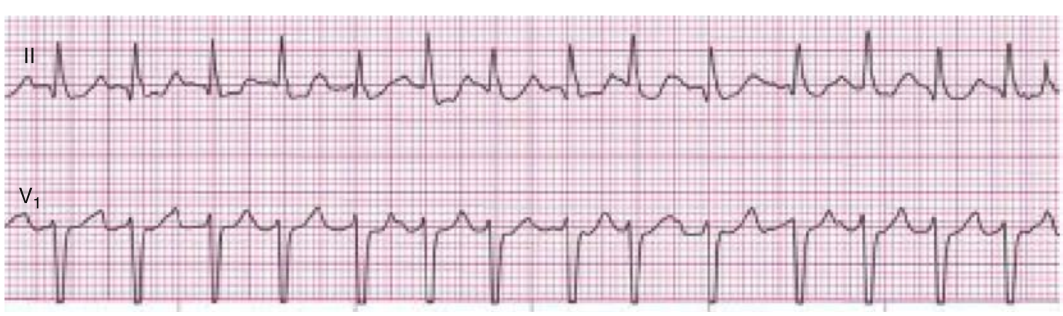

Fig. 10.18

88. **Fig. 10.18**

Rhythm: ___________________ Rate: ___________________ P waves: ____________________________________

PR interval: ________________ QRS duration: __________ QT interval: _________________________________

Interpretation: ____

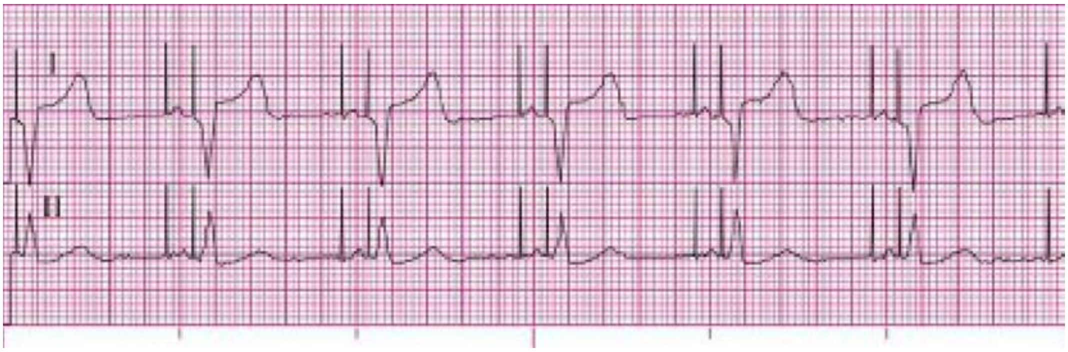

Fig. 10.19

89. Fig. 10.19

Atrial paced activity? ______________ Ventricular paced activity? ____________________________________

Pacemaker malfunction? __________ Identification ____

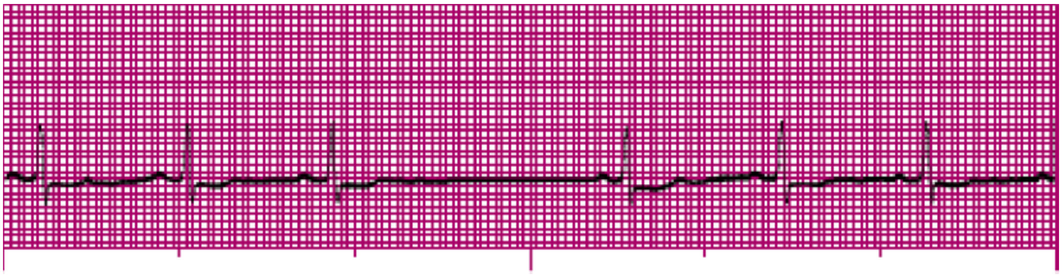

Fig. 10.20

90. Fig. 10.20

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

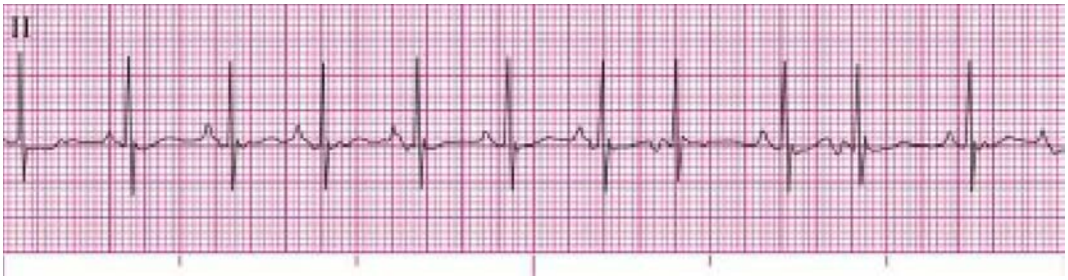

Fig. 10.21 (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

91. Fig. 10.21

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

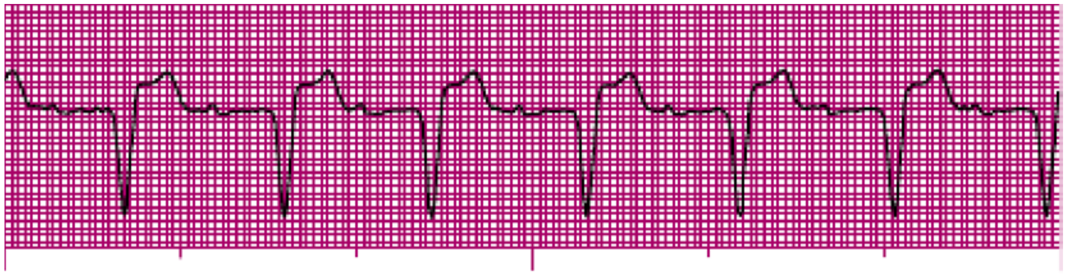

Fig. 10.22 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

92. **Fig. 10.22**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.23 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

93. **Fig. 10.23**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

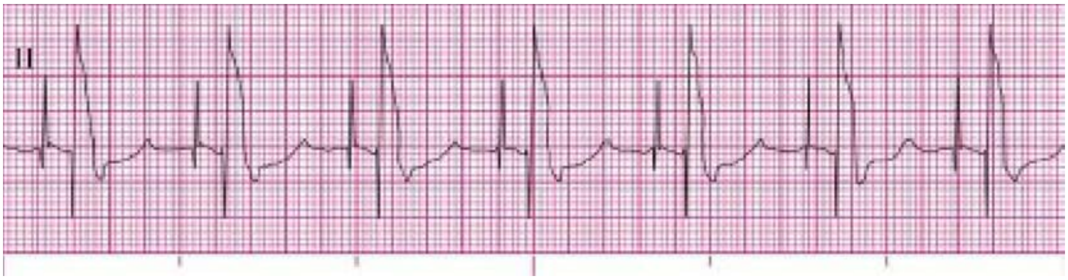

Fig. 10.24 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

94. **Fig. 10.24** This rhythm strip is from a 79-year-old man who experienced a syncopal episode. He has a history of seizures.

Atrial paced activity? ____________ Ventricular paced activity? ________________________________

Pacemaker malfunction? _________ Identification: ____

Fig. 10.25 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

95. **Fig. 10.25** This rhythm strip is from an 81-year-old woman complaining of chest pain.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

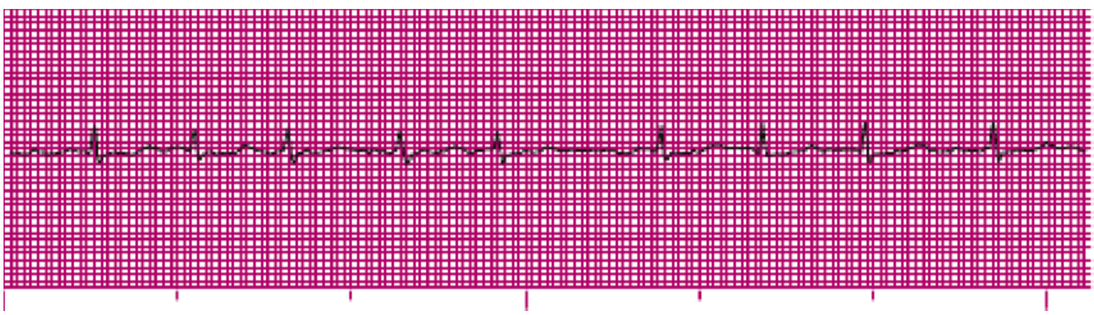

Fig. 10.26

96. **Fig. 10.26** This rhythm strip is from a 74-year-old woman complaining of difficulty breathing.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

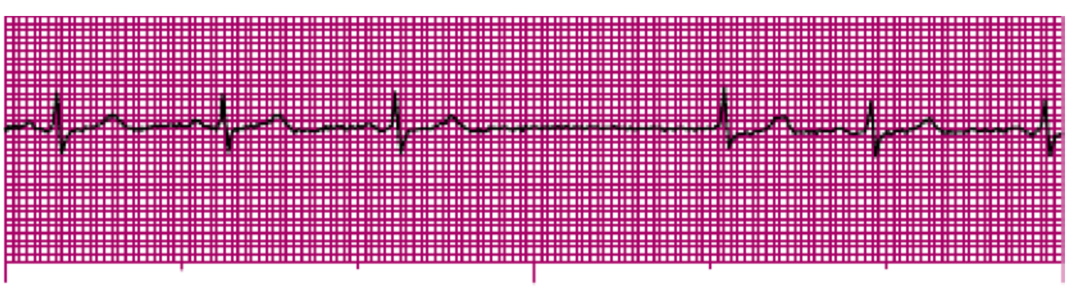

Fig. 10.27

97. **Fig. 10.27** This rhythm strip is from a 63-year-old woman complaining of dizziness.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.28 (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

98. **Fig. 10.28** This rhythm strip is from a 53-year-old man complaining of chest pressure and shortness of breath. He has a history of a spinal cord injury and coronary artery disease.

Rhythm: __________________ Rate: __________________ P waves: ___________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

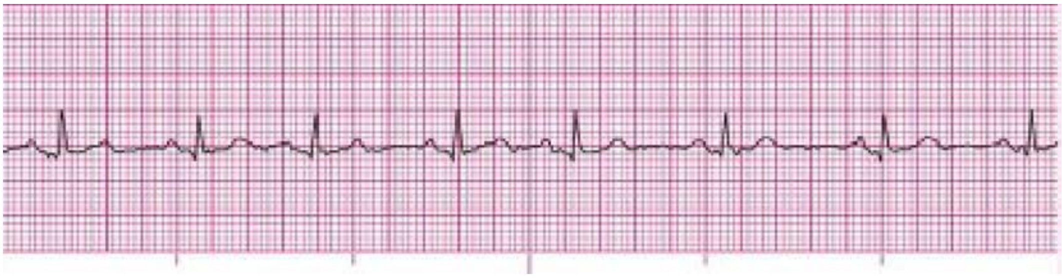

Fig. 10.29

99. **Fig. 10.29** This rhythm strip is from an 82-year-old woman who had a ground-level fall.

Rhythm: __________________ Rate: __________________ P waves: ___________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

Fig. 10.30 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

100. **Fig. 10.30** This rhythm strip is from a 70-year-old man who is complaining of a sharp pain across his shoulders.

Rhythm: __________________ Rate: __________________ P waves: ___________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

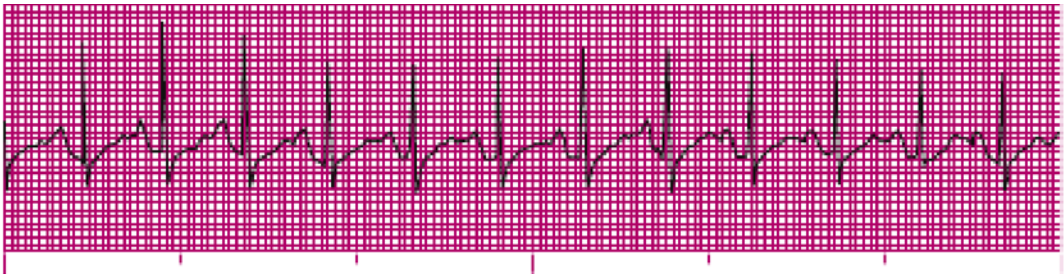

Fig. 10.31 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

101. Fig. 10.31 This rhythm strip is from a 17-year-old man who experienced a syncopal episode while playing baseball in 110°F heat for 4 hours. His core temperature is 101.8°F.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.32 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

102. Fig. 10.32 This rhythm strip is from a 71-year-old man who is complaining of abdominal pain.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.33 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

103. Fig. 10.33

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

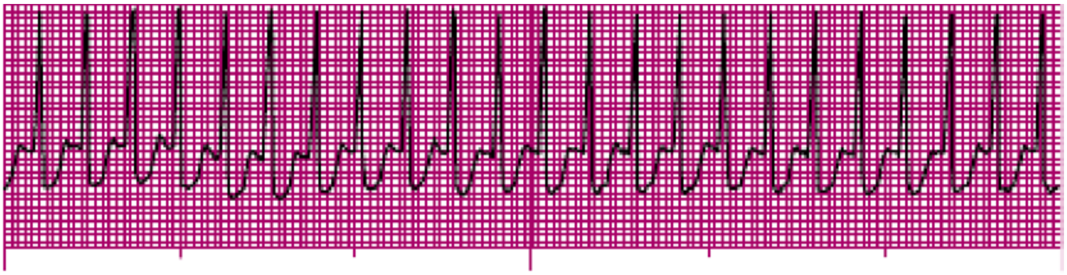

Fig. 10.34 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

104. Fig. 10.34

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

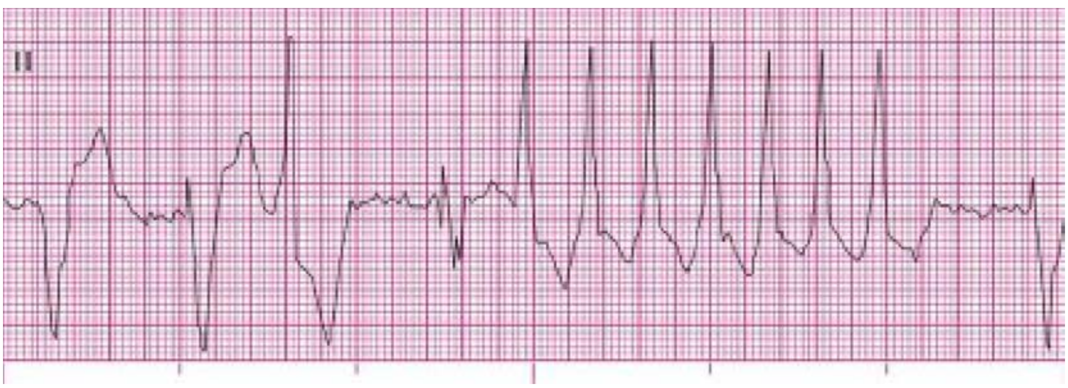

Fig. 10.35 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

105. Fig. 10.35

Atrial paced activity? _________________________ Ventricular paced activity? ________________________

Pacemaker malfunction? ______________________ Identification: ___________________________________

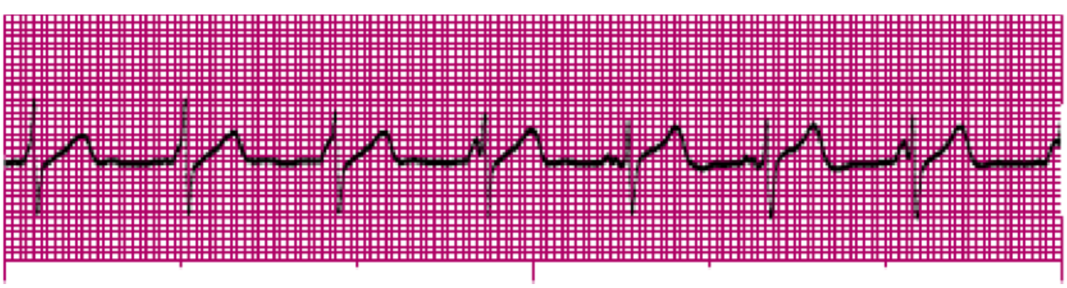

Fig. 10.36

106. Fig. 10.36

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

Fig. 10.37

107. Fig. 10.37

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

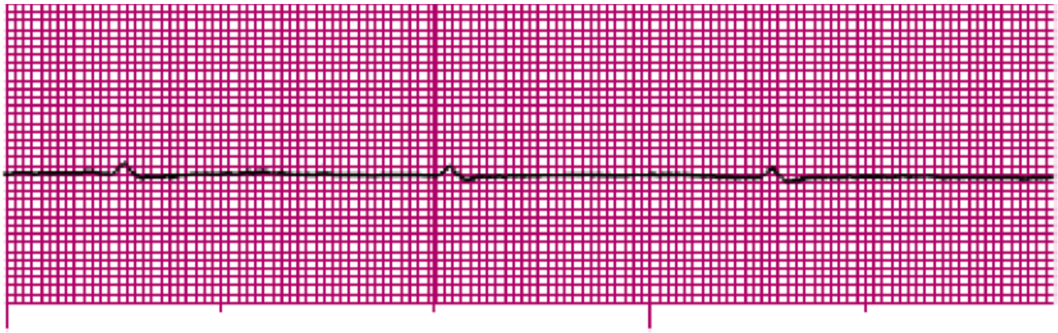

Fig. 10.38 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

108. Fig. 10.38

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

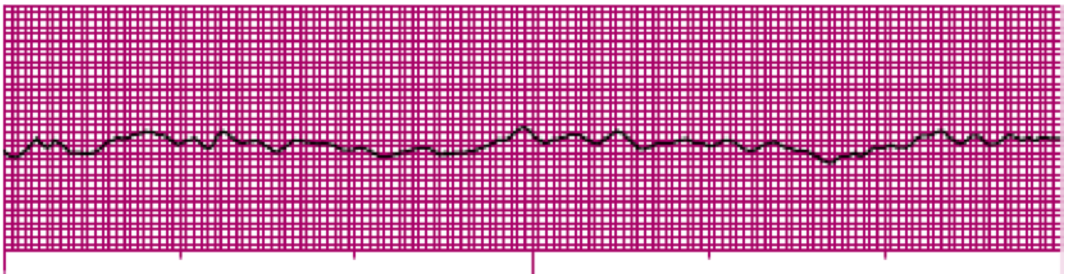

Fig. 10.39 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

109. Fig. 10.39 This rhythm strip is from an 18-year-old man with a gunshot wound to his chest.

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

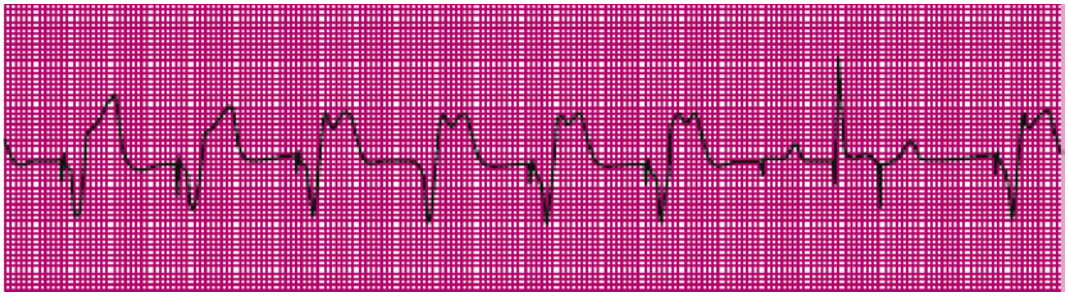

Fig. 10.40 (From Sole ML, Klein DG, Moseley MJ: *Introduction to critical care nursing,* ed 5, Philadelphia, 2008, Saunders.)

110. Fig. 10.40

Atrial paced activity? ______________ Ventricular paced activity? ____________________________________

Pacemaker malfunction? __________ Identification: ____

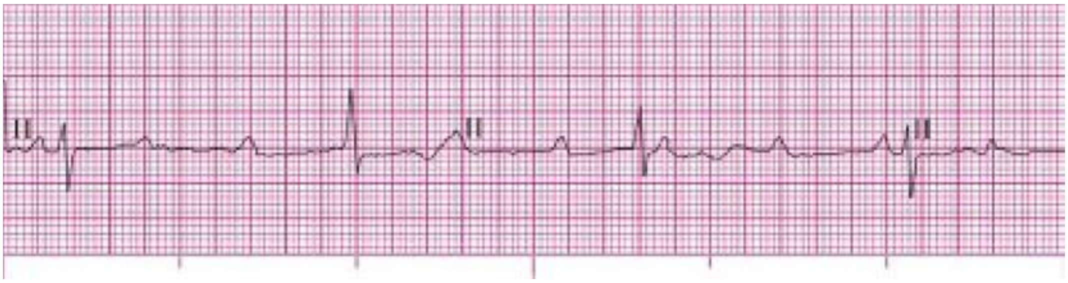

Fig. 10.41

111. Fig. 10.41

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

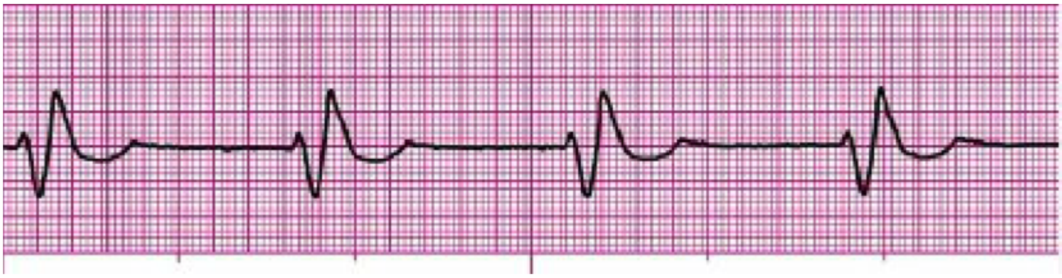

Fig. 10.42 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

112. Fig. 10.42

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

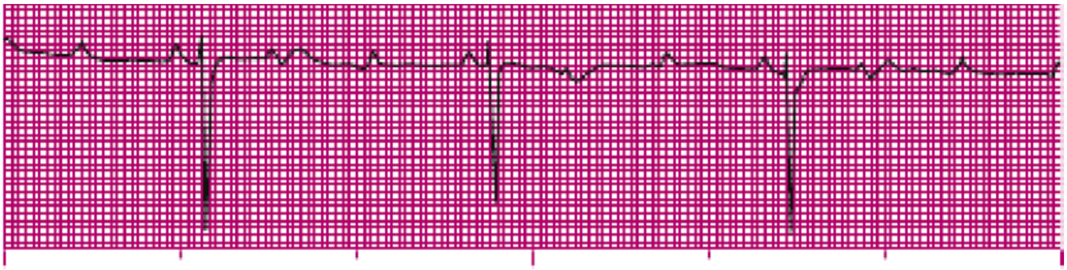

Fig. 10.43 (From Phillips RE, Feeney MK: *The cardiac rhythms: a systematic approach to interpretation,* ed 3, Philadelphia, 1990, Saunders.)

113. **Fig. 10.43**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.44 (From Braunwald E, Libby P, Zipes DP, et al: *Heart disease: a textbook of cardiovascular medicine,* ed 6, St. Louis, 2001, Mosby.)

114. **Fig. 10.44**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

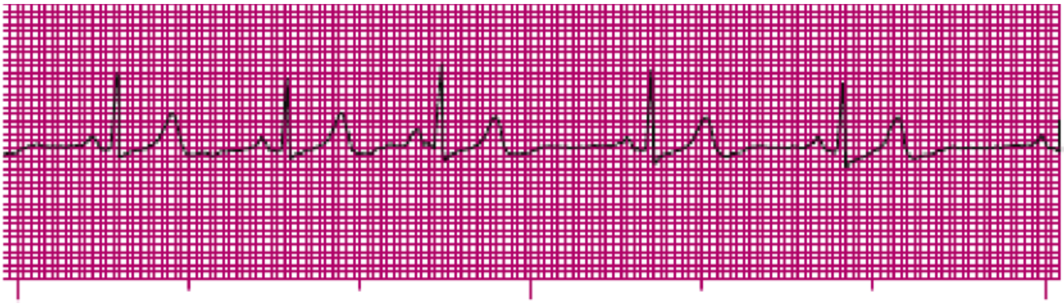

Fig. 10.45 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

115. **Fig. 10.45** This rhythm strip is from an 88-year-old woman complaining of dizziness.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

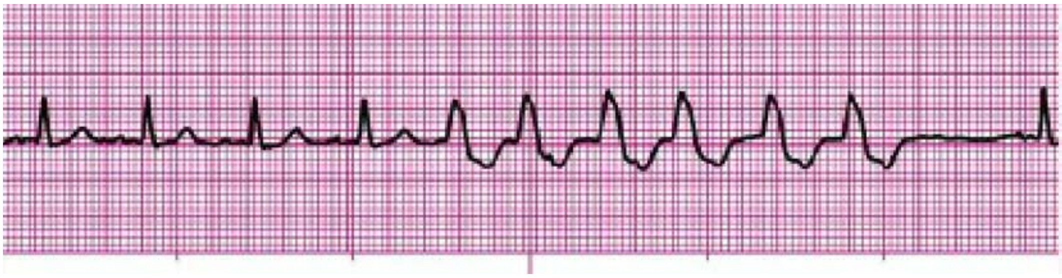

Fig. 10.46 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

116. Fig. 10.46 This rhythm strip is from a 70-year-old man who sustained second-degree burns over 20% of his body. He has a history of diabetes, coronary artery disease, and hypertension.

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

Fig. 10.47

117. Fig. 10.47

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

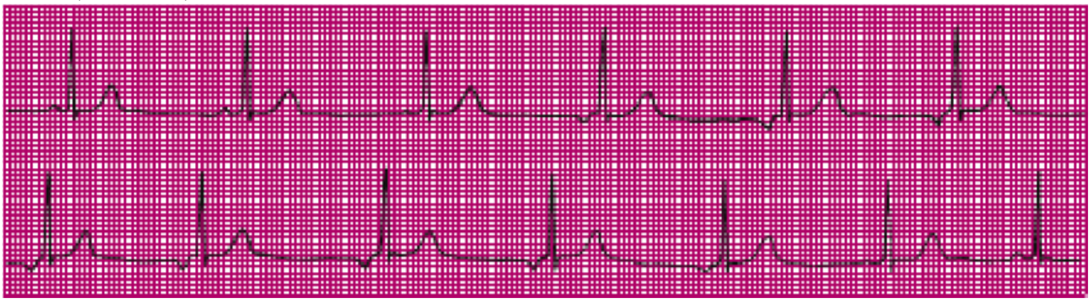

Fig. 10.48 (From Goldberger AL: *Clinical electrocardiography: a simplified approach,* ed 7, St. Louis, 2006, Mosby.)

118. Fig. 10.48

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

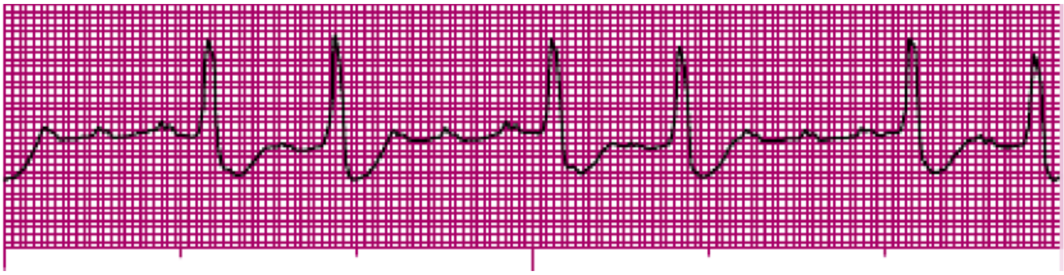

Fig. 10.49

119. Fig. 10.49

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

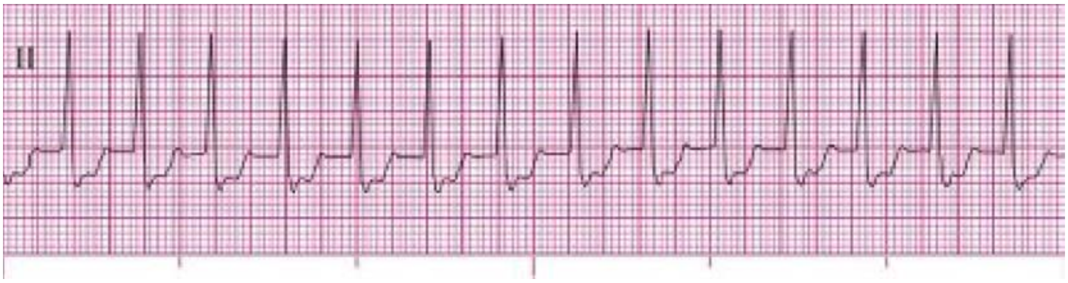

Fig. 10.50

120. Fig. 10.50

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

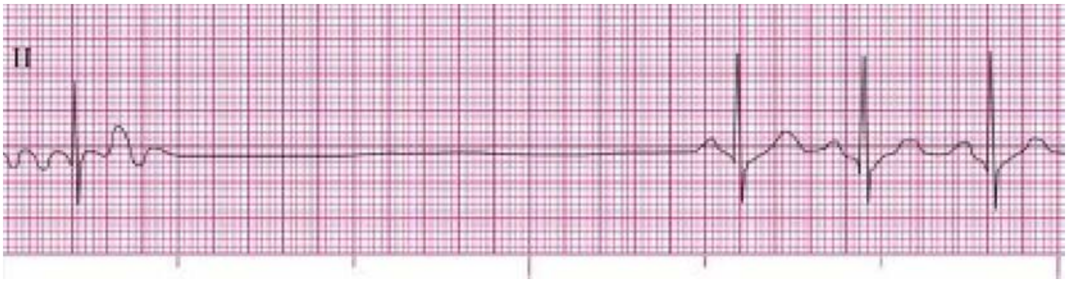

Fig. 10.51 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

121. Fig. 10.51

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

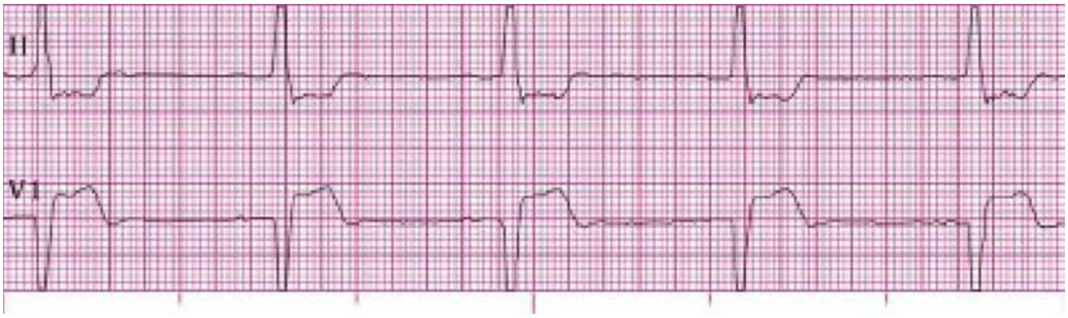

Fig. 10.52

122. Fig. 10.52

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

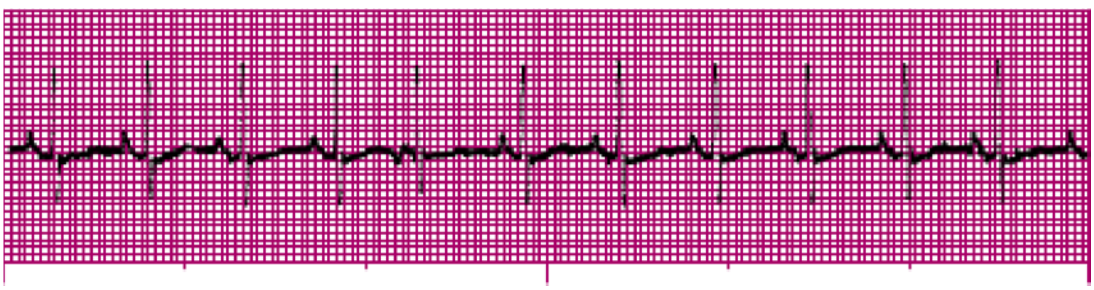

Fig. 10.53

123. Fig. 10.53

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

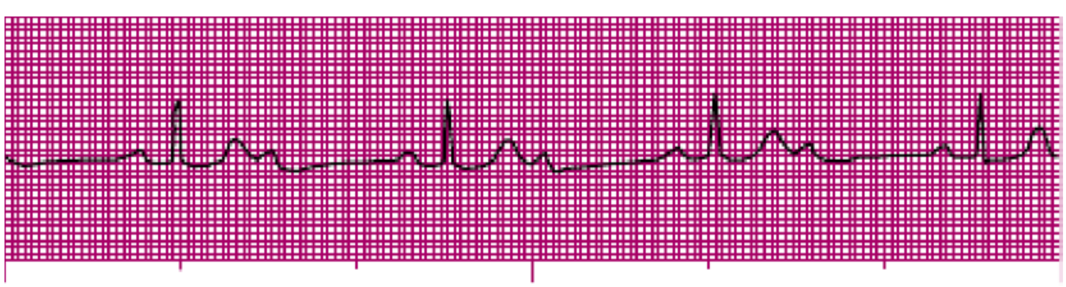

Fig. 10.54 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

124. Fig. 10.54

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.55 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

125. Fig. 10.55

Rhythm: ____________ Rate: ____________ P waves: ____________________

PR interval: ____________ QRS duration: ________ QT interval: ____________________

Interpretation: ____

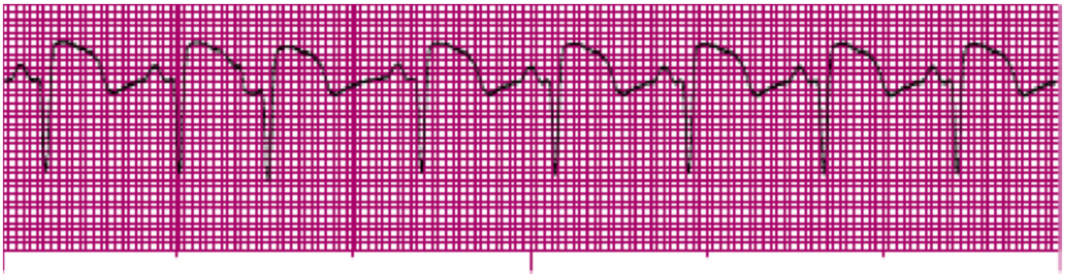

Fig. 10.56

126. Fig. 10.56

Rhythm: ____________ Rate: ____________ P waves: ____________________

PR interval: ____________ QRS duration: ________ QT interval: ____________________

Interpretation: ____

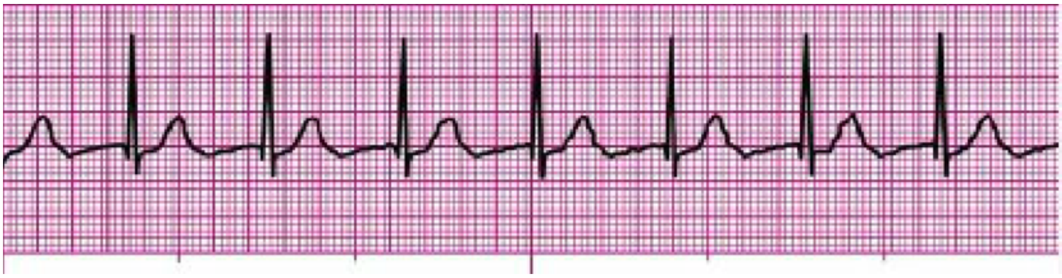

Fig. 10.57 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

127. Fig. 10.57

Rhythm: ____________ Rate: ____________ P waves: ____________________

PR interval: ____________ QRS duration: ________ QT interval: ____________________

Interpretation: ____

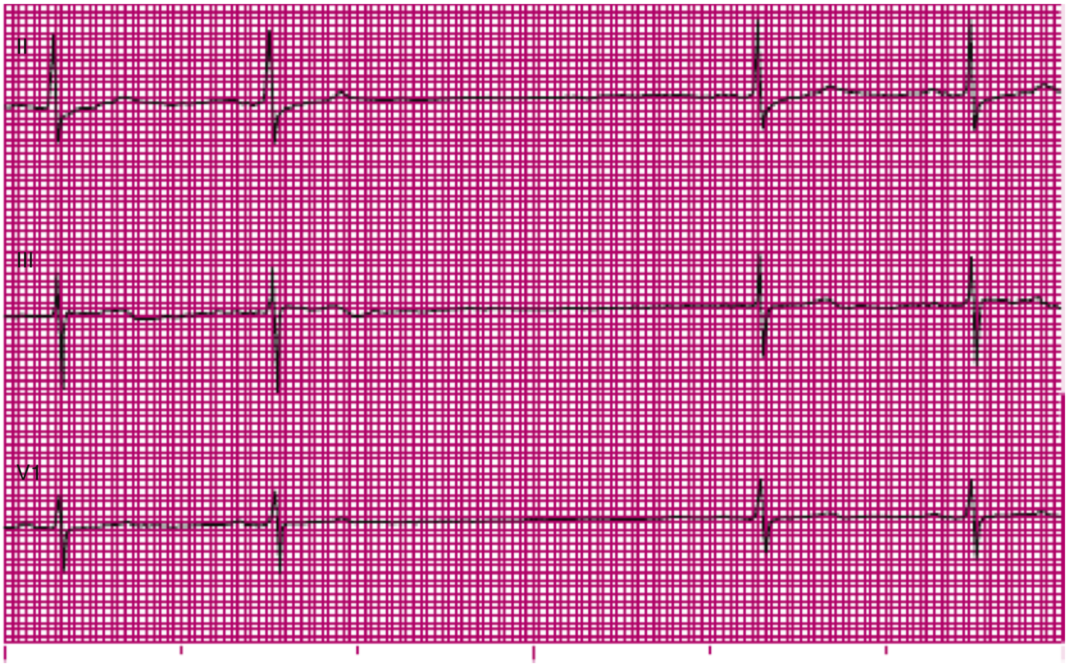

Fig. 10.58

128. Fig. 10.58 These rhythm strips are from a 67-year-old woman complaining of dizziness and chest pain. She has a history of a three-vessel coronary artery bypass graft and hypertension.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

Fig. 10.59 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

129. Fig. 10.59 This rhythm strip is from a 59-year-old man who was driving to work on the freeway when his internal defibrillator discharged. He was asymptomatic at the time this ECG was obtained a few minutes after the event.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

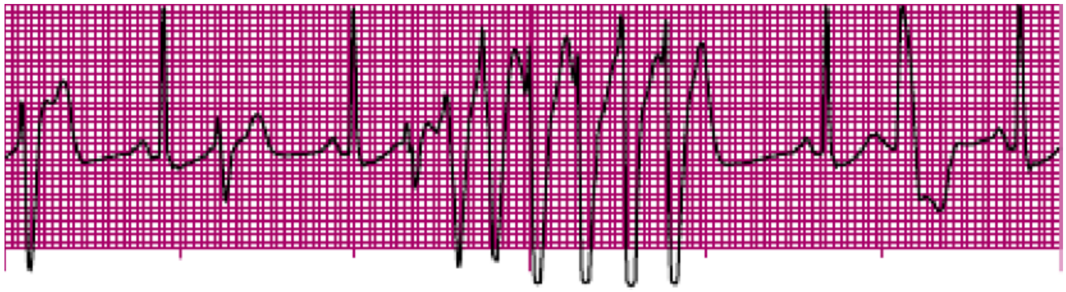

Fig. 10.60 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

130. Fig. 10.60 This rhythm strip is from an 84-year-old man who is complaining of dizziness. He had a triple bypass 4 days ago.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ____________________________

Interpretation: ____

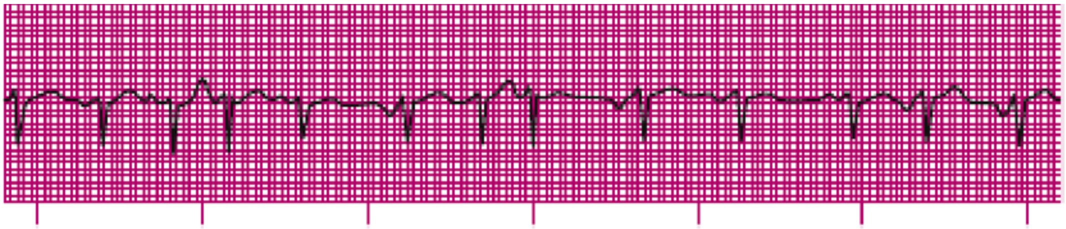

Fig. 10.61 (From Sole ML, Klein DG, Moseley MJ: *Introduction to critical care nursing,* ed 5, Philadelphia, 2008, Saunders.)

131. Fig. 10.61

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ____________________________

Interpretation: ____

Fig. 10.62 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

132. Fig. 10.62

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ____________________________

Interpretation: ____

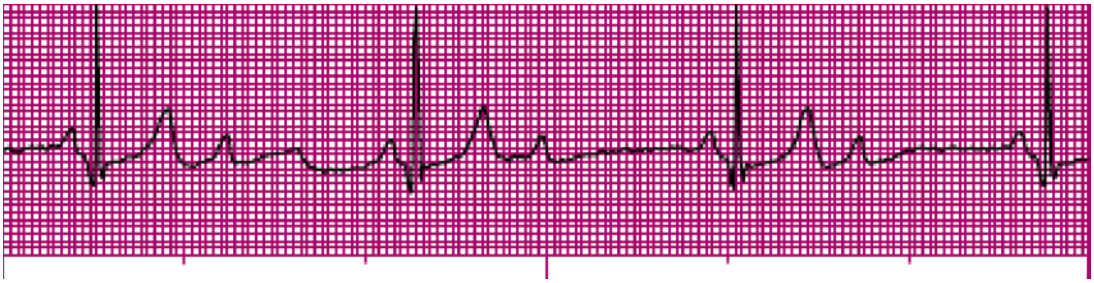

Fig. 10.63 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

133. Fig. 10.63

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

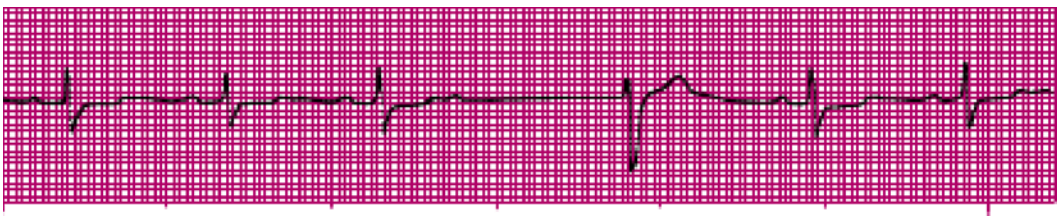

Fig. 10.64 (From Chou T, Ramaiah LS: *Electrocardiography in clinical practice: adult and pediatric,* ed 4, Philadelphia, 1996, Saunders.)

134. Fig. 10.64

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

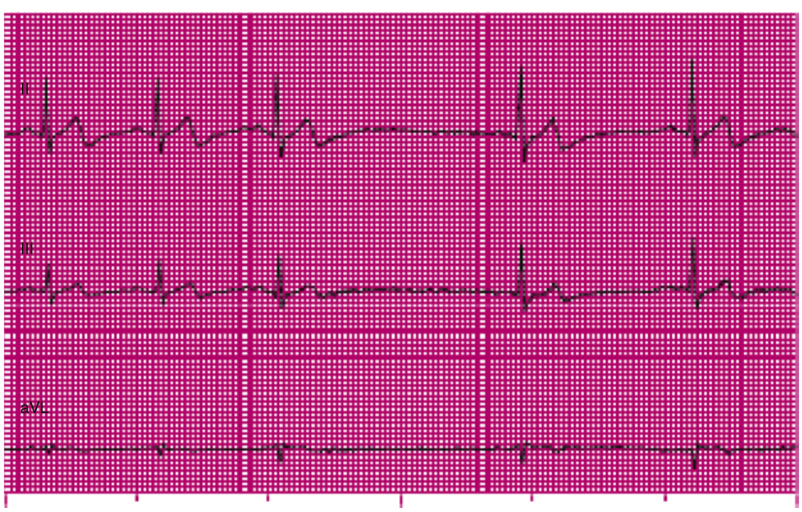

Fig. 10.65

135. Fig. 10.65

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

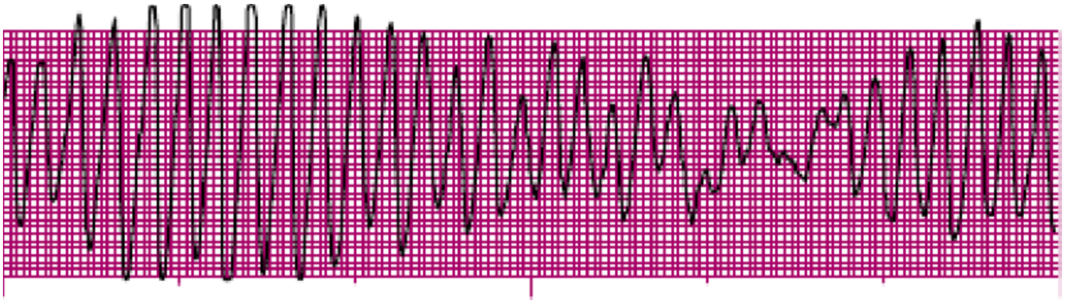

Fig. 10.66 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

136. **Fig. 10.66**

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

Fig. 10.67 (From Andreoli TE, Griggs R, Wing W, Fitz JG: *Andreoli and Carpenter's Cecil essentials of medicine,* ed 9, Philadelphia, 2016, Saunders.)

137. **Fig. 10.67**

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

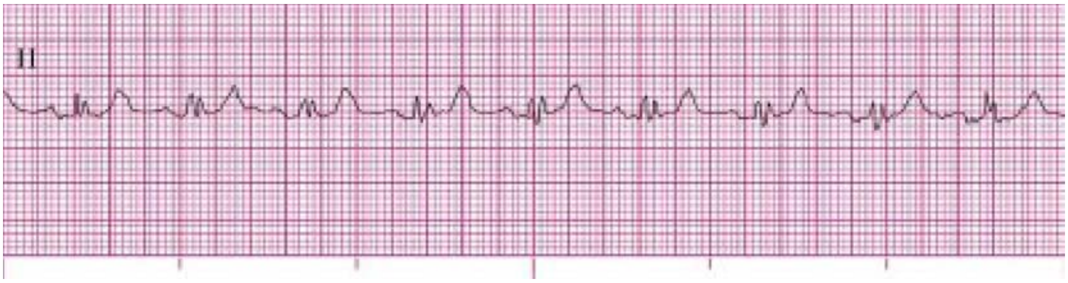

Fig. 10.68 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

138. **Fig. 10.68** This rhythm strip is from a 61-year-old woman complaining of chest pain. She has a history of asthma and chronic obstructive pulmonary disease.

Rhythm: __________________ Rate: __________________ P waves: __________________________________

PR interval: _______________ QRS duration: __________ QT interval: _______________________________

Interpretation: ____

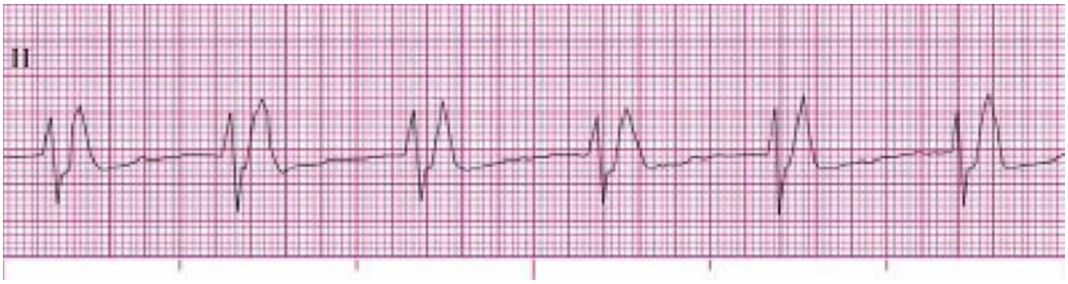

Fig. 10.69 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

139. **Fig. 10.69** This rhythm strip is from a 52-year-old man found unresponsive, apneic, and pulseless.

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

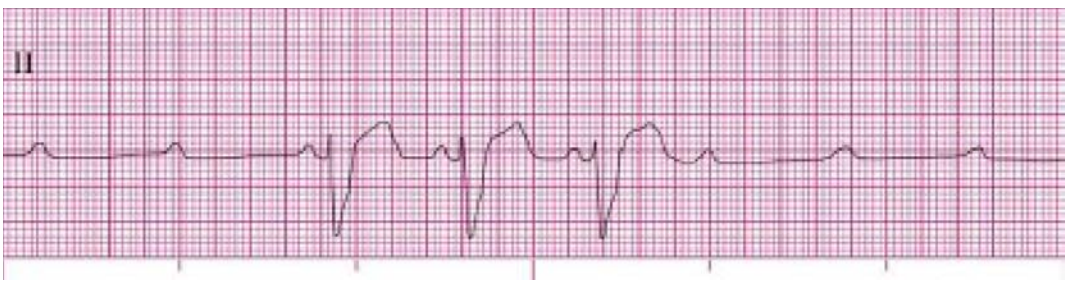

Fig. 10.70

140. **Fig. 10.70**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

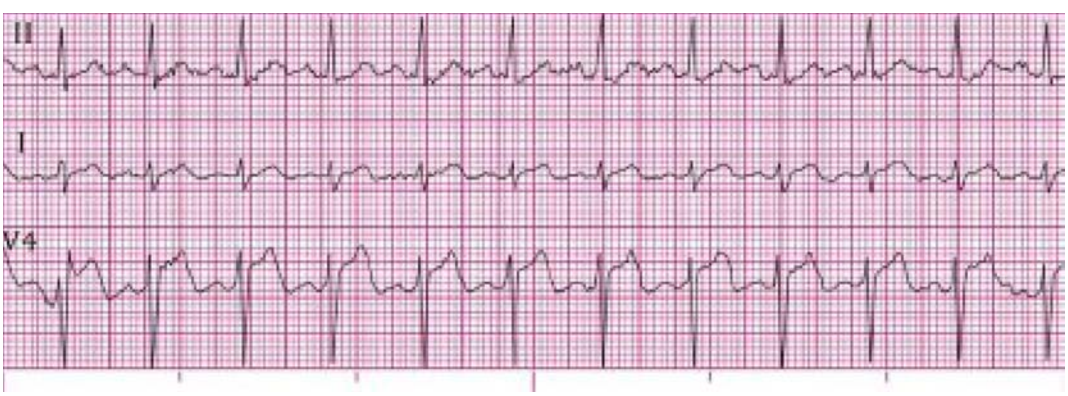

Fig. 10.71

141. **Fig. 10.71**

Rhythm: ________________ Rate: ________________ P waves: ______________________________

PR interval: ______________ QRS duration: _________ QT interval: ___________________________

Interpretation: ____

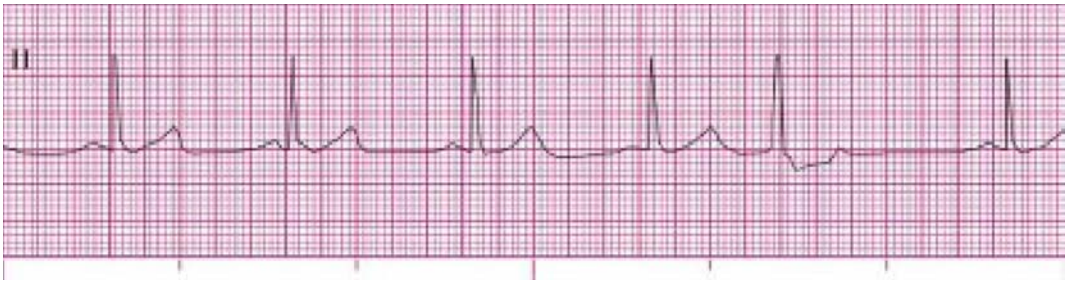

Fig. 10.72 (Modified from Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

142. Fig. 10.72

Rhythm: ______________________ Rate: ______________________ P waves: ______________________

PR interval: __________________ QRS duration: __________________ QT interval: __________________

Interpretation: ____

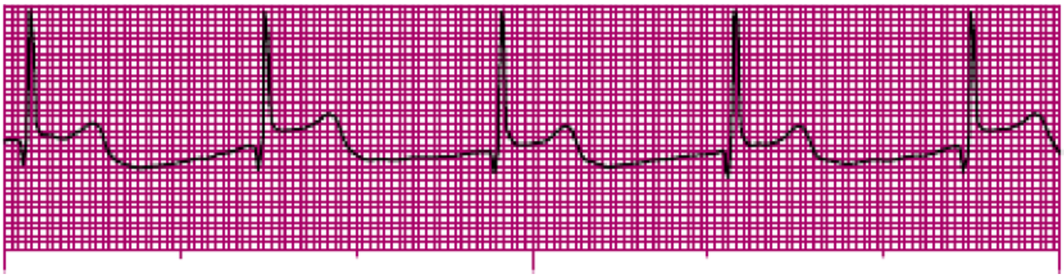

Fig. 10.73 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

143. Fig. 10.73

Rhythm: ______________________ Rate: ______________________ P waves: ______________________

PR interval: __________________ QRS duration: __________________ QT interval: __________________

Interpretation: ____

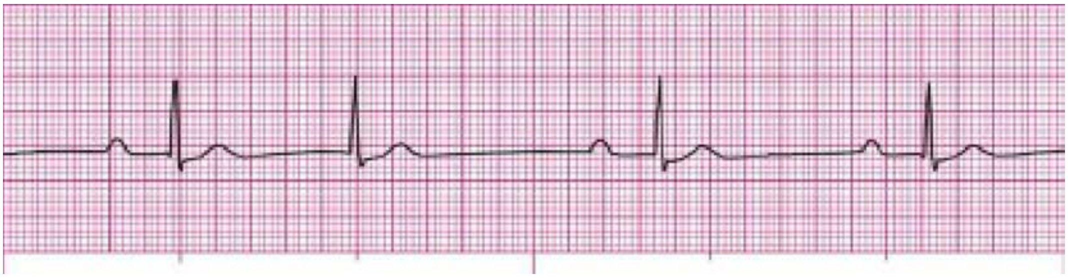

Fig. 10.74 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

144. Fig. 10.74

Rhythm: ______________________ Rate: ______________________ P waves: ______________________

PR interval: __________________ QRS duration: __________________ QT interval: __________________

Interpretation: ____

Fig. 10.75 (From Aehlert B: *ECG study cards,* St. Louis, 2004, Mosby.)

145. Fig. 10.75 These rhythm strips are from an 82-year-old man complaining of back pain.

Rhythm: ______________________ Rate: ______________________ P waves: ____________________

PR interval: ____________________ QRS duration: _______________ QT interval: _________________

Interpretation: ____

Posttest Answers

1. B. A branch of the RCA supplies the right atrium and right ventricle with blood.

OBJ: Name the primary branches and areas of the heart supplied by the right and left coronary arteries.

2. D. Parasympathetic (inhibitory) nerve fibers supply the SA node, atrial muscle, and AV bundle of the heart by the vagus nerves. Parasympathetic stimulation has the following actions:
   - Slows the rate of discharge of the SA node
   - Slows conduction through the AV node
   - Decreases the strength of atrial contraction
   - Can cause a small decrease in the force of ventricular contraction

OBJ: Compare and contrast the effects of sympathetic and parasympathetic stimulation of the heart.

3. B. The flow of blood from the superior and inferior venae cavae into the atria is normally continuous. About 70% of this blood flows directly through the atria and into the ventricles before the atria contract; this is called *passive filling.* When the atria contract, an additional 10% to 30% of the returning blood is added to filling of the ventricles. This additional contribution of blood resulting from atrial contraction is called *atrial kick.* Afterload is the pressure or resistance against which the ventricles must pump to eject blood. Cardiac output is the amount of blood pumped into the aorta each minute by the heart; it is defined as the stroke volume multiplied by the heart rate. Peripheral resistance is the resistance to the flow of blood determined by blood vessel diameter and the tone of the vascular musculature.

OBJ: Explain atrial kick.

4. A. The pulmonic and aortic valves are semilunar valves. The semilunar valves prevent backflow of blood from the aorta and pulmonary arteries into the ventricles. The tricuspid and mitral valves are AV valves, which separate the atria from the ventricles.

OBJ: Name and identify the location of the AV and semilunar valves.

5. D. The left main coronary artery supplies oxygenated blood to its two primary branches: the LAD artery, which is also called the *anterior interventricular artery*, and the Cx artery.

OBJ: Name the primary branches and areas of the heart supplied by the right and left coronary arteries.

6. B. In general, cardiac cells have either a mechanical (i.e., contractile) or an electrical (i.e., pacemaker) function. Pacemaker cells are specialized cells of the electrical conduction system. Pacemaker cells can also be called *conducting cells* or *automatic cells.* They are responsible for the spontaneous generation and conduction of electrical impulses.

OBJ: Describe the two basic types of cardiac cells in the heart, where they are found, and their function.

7. B. During the absolute refractory period, the cell will not respond to further stimulation within itself. This means that the myocardial working cells cannot contract and that the cells of the electrical conduction system cannot conduct an electrical impulse, no matter how strong the internal electrical stimulus. On the ECG, the absolute refractory period begins with the onset of the QRS complex and terminates at approximately the apex of the T wave.

OBJ: Define the absolute, effective, relative refractory, and supernormal periods and their location in the cardiac cycle.

8. D. The QT interval, measured from the beginning of the QRS complex to the end of the T wave, represents the time from initial depolarization of the ventricles to the end of ventricular repolarization.
OBJ: Define and describe the significance of each of the following as they relate to cardiac electrical activity: the P wave, QRS complex, T wave, U wave, PR segment, TP segment, ST segment, PR interval, QRS duration, and QT interval.

9. D. To evaluate the rhythmicity of the atrial rhythm, the interval between two consecutive P waves is measured and compared to succeeding P-P intervals.
OBJ: Describe a systematic approach to the analysis and interpretation of cardiac dysrhythmias.

10. A. A bipolar lead is an ECG lead that has a positive and negative electrode. Each lead records the difference in electrical potential (i.e., voltage) between two selected electrodes. Although all ECG leads are technically bipolar, leads I, II, and III use two distinct electrodes, one of which is connected to the positive input of the ECG machine and the other to the negative input.
OBJ: Describe correct anatomic placement of the standard limb leads, the augmented leads, and the chest leads.

11. A. In sinus arrhythmia, the heart rate increases gradually during inspiration (R-R intervals shorten) and decreases with expiration (R-R intervals lengthen).
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management of sinus arrhythmia.

12. D. The rate of a sinus bradycardia is less than 60 beats/min. R-R and P-P intervals are regular; P waves are positive in lead II, and one precedes each QRS complex. The PR interval is within normal limits, and the QRS duration is 0.11 second or less unless it is abnormally conducted.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management of sinus bradycardia.

13. B. During SA block, also called *sinus exit block*, the pacemaker cells within the SA node initiate an impulse but it is blocked as it exits the SA node; thus, SA block is a disorder of impulse conduction. Sinus arrest, also called *sinus pause* or *SA arrest*, is a disorder of impulse formation. In sinus arrest, the pacemaker cells of the SA node fail to initiate an electrical impulse for one or more beats, resulting in absent PQRST complexes on the ECG.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management of SA block and sinus arrest.

14. D. Signs and symptoms experienced during a tachydysrhythmia are usually primarily related to decreased ventricular filling time and stroke volume. The heart's demand for oxygen increases as the heart rate increases. As the heart rate increases, there is less time for the ventricles to fill and less blood for the ventricles to pump out with each contraction, which can lead to decreased cardiac output. Because the coronary arteries fill when the ventricles are at rest, rapid heart rates decrease the time available for coronary artery filling. This decreases the heart's blood supply. Chest discomfort can result if the supplies of blood and oxygen to the heart are inadequate.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management of sinus tachycardia.

15. B. The terms *ectopic*, which means out of place, or *latent* are used to describe an impulse that originates from a source other than the SA node. Ectopic pacemaker sites include the cells of the AV junction and Purkinje fibers, although their intrinsic rates are slower than that of the SA node.
OBJ: Describe the location, the function, and, when appropriate, the intrinsic rate of the following structures: SA node, AV bundle, and Purkinje fibers.

16. C. *Multiform atrial rhythm* is an updated term for the rhythm formerly known as *wandering atrial pacemaker*. With this rhythm, the size, shape, and direction of the P waves vary, sometimes from beat to beat. The difference in the look of the P waves is a result of the gradual shifting of the dominant pacemaker between the SA node, the atria, and the AV junction. Wandering atrial pacemaker is associated with a normal or slow rate and irregular P-P, R-R, and PR intervals because of the different sites of impulse formation.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for wandering atrial pacemaker (multiform atrial rhythm).

17. D. AVNRT is the most common type of SVT.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for AVNRT.

18. D. Treatment decisions are based on the ventricular rate, the duration of the rhythm, the patient's general health, and how he or she is tolerating the rhythm. It is best to consult a cardiologist when considering specific therapies for AFib. The two primary treatment strategies used to control symptoms associated with AFib are rate control and rhythm control. With rate control, the patient remains in AFib, but the ventricular rate is controlled to decrease acute symptoms, reduce signs of ischemia, and reduce or prevent signs of heart failure from developing. With rhythm control, sinus rhythm is reestablished.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for AFib.

19. C. The term *paroxysmal* is used to describe a rhythm that starts or ends suddenly. Atrial tachycardia that begins or ends suddenly is called PSVT, once called *paroxysmal atrial tachycardia* (PAT). PSVT may last for minutes, hours, or days. If the onset or end of PSVT is not observed on the ECG, the dysrhythmia is simply called *SVT.*

OBJ: Explain the terms *PAT* and *PSVT.*

20. A. AFib is the most common sustained dysrhythmia in adults, and it occurs because of altered automaticity in one or several rapidly firing sites in the atria or reentry involving one or more circuits in the atria.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for AFib.

21. D. You can usually tell the difference between a PAC and a PJC by the P wave. A PAC typically has an upright P wave before the QRS complex in leads II, III, and aVF. A P wave may or may not be present with a PJC. If a P wave is present, it is inverted (retrograde) and may precede or follow the QRS. PJCs can be misdiagnosed when the P wave of a PAC is buried in the preceding T wave.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for PJCs.

22. B. Characteristic ECG findings with a WPW pattern include a short PR interval, QRS widening, and a delta wave. A delta wave is an initial slurred deflection at the beginning of the QRS complex that results from the initial activation of the QRS by conduction over the accessory pathway.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for AVNRT.

23. D. In atrial flutter, atrial waveforms are produced that resemble the teeth of a saw or the boards of a picket fence; these are called *flutter waves*, which are best observed in leads II, III, aVF, and V_1.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for atrial flutter.

24. B. If the AV junction paces the heart, the electrical impulse must travel in a backward (retrograde) direction to activate the atria. If the atria depolarize before the ventricles, an inverted P wave will be seen *before* the QRS complex, and the PR interval will usually measure 0.12 second or less. The PR interval is shorter than usual because an impulse that begins in the AV junction does not have to travel as far to stimulate the ventricles. If the atria and ventricles depolarize at the same time, a P wave will not be visible because it will be hidden in the QRS complex. When the atria are depolarized after the ventricles, the P wave typically distorts the end of the QRS complex, and an inverted P wave will appear *after* the QRS.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for a junctional escape rhythm.

25. C. Nonparoxysmal (i.e., gradual onset) junctional tachycardia usually starts as an accelerated junctional rhythm, but the heart rate gradually increases to more than 100 beats/min. The usual ventricular rate for nonparoxysmal junctional tachycardia is 101 to 140 beats/min. Paroxysmal junctional tachycardia, which is also known as focal or automatic junctional tachycardia, is an uncommon dysrhythmia that starts and ends suddenly and that is often precipitated by a PJC. The ventricular rate for paroxysmal junctional tachycardia is generally faster, at a rate of 140 beats/min or more.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for junctional tachycardia.

26. B. Junctional (or ventricular) complexes may come early (before the next expected sinus beat) or late (after the next expected sinus beat). If the complex is *early,* it is called a <u>premature complex</u>—more specifically, a premature junctional (or ventricular) complex. If the complex is *late,* it is called a junctional (or ventricular) <u>escape beat</u>. To determine if a complex is early or late, you need to see at least two sinus beats in a row to establish the regularity of the underlying rhythm.

OBJ: Explain the difference between PJCs and junctional escape beats.

27. B. The term *symptomatic bradycardia* is used to describe a patient who experiences signs and symptoms of hemodynamic compromise related to a slow heart rate. Treatment of a symptomatic bradycardia should include assessing the patient's oxygen saturation level and determining if signs of increased work of breathing are present (e.g., retractions, tachypnea, paradoxic abdominal breathing). Give supplemental oxygen if oxygenation is inadequate. Assist breathing if ventilation is inadequate, establish intravenous access, and obtain a 12-lead ECG. Intravenous administration of atropine, the drug of choice for symptomatic bradycardia, may be necessary if the patient's symptoms warrant it. Reassess the patient's response and continue monitoring him or her.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for a junctional escape rhythm.

28. C. Three or more sequential PVCs are termed a *run* or *burst*, and three or more PVCs that occur in a row at a rate of more than 100 beats/min are considered a run of VT.

OBJ: Explain the terms *bigeminy*, *trigeminy*, *quadrigeminy*, and *run* when used to describe premature complexes.

29. A. PVCs that look alike in the same lead and begin from the same anatomic site (i.e., focus) are called <u>uniform</u> *PVCs.*

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for PVCs.

30. C. PMVT is characterized by QRS complexes that vary in shape and amplitude from beat to beat and appear to twist from upright to negative or negative to upright and back, resembling a spindle. The ventricular rate is 150 to 300 beats/min and is typically 200 to 250 beats/min.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for PMVT.

31. D. When a delay or interruption in impulse conduction from the atria to the ventricles occurs as a result of a transient or permanent anatomic or functional impairment, the resulting dysrhythmia is called an *AV block*. A BBB is a disruption in impulse conduction from the bundle of His through either the right or left bundle branch to the Purkinje fibers. With SA block, which is also called *sinus exit block*, the pacemaker cells within the SA node initiate an impulse but it is blocked as it exits the SA node. With sinus arrest, the pacemaker cells of the SA node fail to initiate an electrical impulse for one or more beats, resulting in absent PQRST complexes on the ECG.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management for first-degree AV block.

32. B. The rSR′ pattern is characteristic of RBBB. The rSR′ pattern is sometimes referred to as an "M" or "rabbit ear" pattern.

OBJ: Describe the appearance of RBBB and LBBB as seen in lead V_1.

33. B. A first-degree AV block is present when there is a 1:1 relationship between P waves and QRS complexes and the PR interval is constant and more than 0.20 second in duration (prolonged).

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management for first-degree AV block.

34. D. Second-degree AV blocks are types of *incomplete* blocks because at least some of the impulses from the SA node are conducted to the ventricles. With third-degree AV block, there is a *complete* block in conduction of impulses between the atria and the ventricles.

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management for third-degree AV block.

35. B. Second-degree 2:1 AV block is characterized by P waves that are normal in size and shape, but every other P wave is not followed by a QRS. The atrial rate is twice the ventricular rate. Because there are no two PQRST cycles in a row from which to compare PR intervals, 2:1 AV block cannot be conclusively classified as type I or type II. To determine the type of block with certainty, it is necessary to continue close ECG monitoring of the patient until the conduction ratio of P waves to QRS complexes changes to 3:2, 4:3, and so on, which would enable PR interval comparison. With second-degree AV block in the form of 2:1 AV block, the level of the block can be located within the AV node or within the His-Purkinje system.

OBJ: Describe 2:1 AV block and advanced second-degree AV block.

36. B. The terms *advanced* or *high-grade second-degree AV block* may be used to describe three or more consecutive P waves that are not conducted. For example, with 3:1 AV block, every third P wave is conducted (i.e., followed by a QRS complex); with 4:1 AV block, every fourth P wave is conducted.

OBJ: Describe 2:1 AV block and advanced second-degree AV block.

37. A. When the presence of BBB is suspected, an examination of V_1 can reveal whether the block affects the right or the left bundle branch.

OBJ: Describe the appearance of RBBB and LBBB as seen in lead V_1.

38. D. *Capture* refers to the successful conduction of an artificial pacemaker's impulse through the myocardium, resulting in depolarization. A pacemaker spike is a vertical line on the ECG that indicates the pacemaker has discharged. Sensitivity is the extent to which an artificial pacemaker recognizes intrinsic cardiac electrical activity. Inhibition is a pacemaker response in which the output pulse is suppressed when an intrinsic event is sensed.

OBJ: Discuss the terms *triggering*, *inhibition*, *pacing*, *capture*, *electrical capture*, *mechanical capture*, and *sensitivity*.

39. B. The 12-lead ECG provides a 2.5-second view of each lead because it is assumed that 2.5 seconds is long enough to capture at least one representative complex. However, a 2.5-second view is not long enough to properly assess rate and rhythm, so at least one continuous rhythm strip is usually included at the bottom of the tracing.

OBJ: Describe a systematic method for analyzing a 12-lead ECG.

40. C. Although a right ventricular infarction may occur by itself, it is more commonly associated with an inferior MI, and it should be suspected when ECG changes suggesting an inferior infarction are seen.

OBJ: Recognize the changes on the ECG that may reflect evidence of myocardial ischemia, injury, or infarction.

41. B. *Poor R-wave progression* is a phrase used to describe R waves that decrease in size from V_1 to V_4. This is often seen in an anteroseptal infarction but may be a normal variant in young people, particularly in young women. Other causes of poor R-wave progression include LBBB, left ventricular hypertrophy, and severe chronic obstructive pulmonary disease (particularly emphysema).

OBJ: Distinguish patterns of normal and abnormal R-wave progression.

42. The right atrium receives deoxygenated blood from the superior vena cava (which carries blood from the head and upper extremities), the inferior vena cava (which carries blood from the lower body), and the coronary sinus (which receives blood from the intracardiac circulation).

OBJ: Identify and describe the chambers of the heart and the vessels that enter or leave each.

43. A beat originating from the AV junction that appears later than the next expected sinus beat is called a junctional escape beat.
OBJ: Describe the ECG characteristics and possible causes for junctional escape beats.

44. A rapid, wide-QRS rhythm associated with pulselessness, shock, or heart failure should be presumed to be ventricular tachycardia.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for monomorphic VT.

45. PACs associated with a wide QRS complex are called aberrantly conducted PACs, indicating conduction through the ventricles is abnormal.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for PACs.

46. Diastole is the period of relaxation during which a heart chamber is filling.
OBJ: Identify and discuss each phase of the cardiac cycle.

47. The thick, muscular middle layer of the heart wall that contains the atrial and ventricular muscle fibers necessary for contraction is the myocardium.
OBJ: Identify the three cardiac muscle layers.

48. Delivery of an electrical current timed for delivery during the QRS complex is called synchronized cardioversion.
OBJ: Discuss the indications and procedure for synchronized cardioversion.

49. Sometimes, when a PAC occurs very prematurely and close to the T wave of the preceding beat, only a P wave may be seen with no QRS after it (appearing as a pause). This type of PAC is termed a nonconducted (or *blocked*) *PAC*.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for PACs.

50. If the AV junction paces the heart, the electrical impulse must travel in a backward (retrograde) direction to activate the atria.
OBJ: Describe the location, the function, and, when appropriate, the intrinsic rate of the following structures: the SA node, the AV bundle, and the Purkinje fibers.

51. A demand pacemaker is also known as a synchronous or noncompetitive pacemaker.
OBJ: Explain the differences between single-chamber and dual-chamber pacemakers and between fixed-rate and demand pacemakers.

52. The axes of leads I, II, and III form an equilateral triangle with the heart at the center (Einthoven's triangle). If the augmented limb leads are added to this configuration and the axes of the six leads move in a way in which they bisect each other, the result is the hexaxial reference system.
OBJ: Explain the term *electrical axis* and its significance.

53.

Leads	**Heart Surface Viewed**
II, III, aVF	Inferior
V_1, V_2	Septal
V_3, V_4	Anterior
I, aVL, V_5, V_6	Lateral

OBJ: Relate the cardiac surfaces or areas represented by the ECG leads.

54. The pathway of blood flow through the normal heart and lungs to the systemic circulation:

1	Right atrium	9	Left ventricle
8	Mitral valve	5	Pulmonary arteries
11	Aorta	2	Tricuspid valve
3	Right ventricle	6	Pulmonary veins
4	Pulmonic valve	10	Aortic valve
7	Left atrium	12	Systemic circulation

OBJ: Beginning with the right atrium, describe blood flow through the normal heart and lungs to the systemic circulation.

55.
1. Assess regularity (atrial and ventricular)
2. Assess rate (atrial and ventricular)
3. Identify and examine waveforms
4. Assess intervals (e.g., PR, QRS, QT) and examine ST segments
5. Interpret the rhythm and assess its clinical significance

OBJ: Describe a systematic approach to the analysis and interpretation of cardiac dysrhythmias.

56. A dual-chamber pacemaker stimulates the right atrium and right ventricle sequentially (stimulating first the atrium and then the ventricle), mimicking normal cardiac physiology and thus preserving the atrial contribution to ventricular filling (atrial kick).
OBJ: Explain the differences between single-chamber and dual-chamber pacemakers and between fixed-rate and demand pacemakers.

57. ECG monitoring may be used to (1) monitor a patient's heart rate; (2) evaluate the effects of disease or injury on heart function; (3) evaluate pacemaker function; (4) evaluate the response to medications (e.g., antiarrhythmics); (5) obtain a baseline recording before, during, and after a medical procedure; and (6) evaluate for signs of myocardial ischemia, injury, and infarction.
OBJ: Explain the purpose of ECG monitoring.

58. An abnormal (i.e., pathologic) Q wave is more than 0.04 second in duration or more than one third the height of the following R wave in that lead. MI is one possible cause of abnormal Q waves.
OBJ: Define and describe the significance of each of the following as they relate to cardiac electrical activity: P wave, QRS complex, T wave, U wave, PR segment, TP segment, ST segment, PR interval, QRS duration, and QT interval.

59. A sinus rhythm has a rate of 60 to 100 beats/min. A sinus tachycardia has a rate of 101 to 180 beats/min.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and emergency management of sinus tachycardia.

60. Artifact may be caused by loose electrodes, broken wires or ECG cables, muscle tremor, patient movement, external chest compressions, or 60-cycle interference.
OBJ: Define the term *artifact* and explain methods that may be used to minimize its occurrence.

61. Coarse VF is 3 mm or more in amplitude. Fine VF is less than 3 mm in amplitude.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for VF.

62. The AV junction may assume responsibility for pacing the heart under any of the following circumstances:

1. The SA node fails to discharge (such as sinus arrest).
2. An impulse from the SA node is generated but blocked as it exits the SA node (e.g., SA block).
3. The rate of discharge of the SA node is slower than that of the AV junction (e.g., a sinus bradycardia or the slower phase of a sinus arrhythmia).
4. An impulse from the SA node is generated and is conducted through the atria but is not conducted to the ventricles (e.g., an AV block).

OBJ: Describe the location, the function, and, when appropriate, the intrinsic rate of the following structures: the SA node, the AV bundle, and the Purkinje fibers.

63. Because the atria do not contract effectively and expel all of the blood within them, blood may pool within them and form clots. A clot may dislodge on its own or because of conversion to a sinus rhythm. A stroke can result if a clot moves from the atria and lodges in an artery in the brain.
OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for AFib.

64. On the ECG, the ST segment represents early ventricular repolarization, and the T wave represents ventricular repolarization.
OBJ: Define and describe the significance of each of the following as they relate to cardiac electrical activity: P wave, QRS complex, T wave, U wave, PR segment, TP segment, ST segment, PR interval, QRS duration, and QT interval.

65.

ECG Finding	AVNRT	Atrial Flutter	Atrial Fibrillation
Rhythm	**Ventricular rhythm is usually very regular**	**Atrial rhythm is regular; ventricular rhythm is regular or irregular**	**Ventricular rhythm is usually irregularly irregular**
Rate (beats/min)	**150 to 250**	**Atrial rate is typically 240 to 300; ventricular rate variable—determined by AV blockade**	**Atrial rate is 300 to 600; ventricular rate is variable**
P waves (lead II)	**P waves often hidden in QRS complex**	**No identifiable P waves; saw-tooth flutter waves present**	**No identifiable P waves; fibrillatory waves present; erratic, wavy baseline**
PR interval	**If P waves are seen, the PRI will usually measure 0.12 to 0.20 sec**	**Not measurable**	**Not measurable**
QRS duration	**0.11 sec or less unless abnormally conducted**	**0.11 sec or less unless abnormally conducted**	**0.11 sec or less unless abnormally conducted**

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for AVNRT, atrial flutter, and AFib.

66. During pacing, the cardiac monitor is observed for electrical capture, usually evidenced by a wide QRS and broad T wave. In some patients, electrical capture is less obvious, indicated only as a change in the shape of the QRS. Mechanical capture is evaluated by assessing the patient's right upper extremity or right femoral pulses.
OBJ: Discuss the terms *triggering*, *inhibition*, *pacing*, *capture*, *electrical capture*, *mechanical capture*, and *sensitivity*.

67. The four primary characteristics of cardiac cells are listed below:

1. Automaticity
2. Excitability (i.e., irritability)
3. Conductivity
4. Contractility

OBJ: Describe the primary characteristics of cardiac cells.

68. The term *anatomically contiguous leads* refers to leads that "see" the same area of the heart. Two leads are contiguous if they look at the same or adjacent areas of the heart or if they are numerically consecutive *chest* leads.
OBJ: Relate the cardiac surfaces or areas represented by the ECG leads.

69.

	Second-Degree AV Block Type II	2:1 AV Block
Ventricular rhythm	**Irregular**	**Regular**
PR interval	**Constant**	**Constant**
QRS width	**Narrow or wide**	**Narrow or wide**

OBJ: Describe 2:1 AV block and advanced second-degree AV block.

70.

ECG Finding	Idioventricular Rhythm	Accelerated Idioventricular Rhythm	Monomorphic Ventricular Tachycardia
Rhythm	**Essentially regular**	**Essentially regular**	**Usually regular**
Rate (beats/min)	**20 to 40**	**41 to 100; some experts consider the rate 41 to 120**	**101 to 250; some experts consider the rate 121 to 250**
P waves (lead II)	**Usually absent**	**Usually absent**	**Usually absent**
PR interval	**None**	**None**	**None**
QRS duration	**0.12 sec or greater**	**0.12 sec or greater**	**0.12 sec or greater**

OBJ: Describe the ECG characteristics, possible causes, signs and symptoms, and initial emergency care for an idioventricular rhythm, accelerated idioventricular rhythm, and monomorphic VT.

Posttest Rhythm Strip Answers

Note: Because of the distortion of ECGs that can occur during printing, a range of acceptable measurements is provided in the rhythm strip answers throughout this textbook.

71. **Fig. 10.1**
Rhythm: Regular
Rate: 88 beats/min
P waves: Inverted before each QRS; 1:1 relationship
PR interval: 0.12 to 0.14 second
QRS duration: 0.06 second
QT interval: 0.44 to 0.48 second
Interpretation: Accelerated junctional rhythm at 88 beats/min

72. **Fig. 10.2**
Rhythm: Irregular
Rate: 80 beats/min
P waves: Upright before each QRS; an early P wave distorts the T wave of beat 3
PR interval: 0.12 to 0.14 second
QRS duration: 0.06 to 0.08 second
QT interval: 0.36 second
Interpretation: Sinus rhythm at 80 beats/min with a nonconducted PAC and ST-segment depression

73. **Fig. 10.3**
Rhythm: Ventricular irregular; atrial regular
Rate: Ventricular about 70 beats/min; atrial 107 beats/min
P waves: Upright; more Ps than QRSs
PR interval: Lengthening
QRS duration: 0.08 second
QT interval: 0.32 second
Interpretation: Second-degree AV block type 1 at 70 beats/min

74. **Fig. 10.4**
Rhythm: Irregular
Rate: 90 beats/min
P waves: Upright before most QRS complexes; none visible for 3 early beats
PR interval: 0.16 second (sinus beats)
QRS duration: 0.06 second (sinus beats)
QT interval: 0.32 second (sinus beats)
Interpretation: Sinus rhythm at 90 beats/min with ventricular trigeminy

75. **Fig. 10.5**
Rhythm: Irregular
Rate: 70 beats/min
P waves: None visible; fibrillatory waves present
PR interval: None
QRS duration: 0.08 second (atrial beats)
QT interval: Unable to determine
Interpretation: AFib at 70 beats/min with a ventricular complex

76. **Fig. 10.6**
Rhythm: Regular
Rate: 88 beats/min
P waves: Upright before each QRS; 1:1 relationship
PR interval: 0.16 second
QRS duration: 0.06 to 0.08 second
QT interval: 0.32 to 0.36 second
Interpretation: Sinus rhythm at 88 beats/min; artifact is present

77. **Fig. 10.7**
Rhythm: Irregular
Rate: 30 beats/min (junctional beats) to 56 beats/min (sinus beats)
P waves: None with junctional beats; upright with sinus beats

PR interval: None with junctional beats; 0.18 second with sinus beats
QRS duration: 0.04 second (junctional beats); 0.08 second (sinus beats)
QT interval: 0.36 second
Interpretation: Junctional bradycardia at 30 beats/min to sinus bradycardia at 56 beats/min

78. **Fig. 10.8**
Rhythm: Regular
Rate: 65 beats/min
P waves: Upright before each QRS, some are notched; 1:1 relationship
PR interval: 0.16 second
QRS duration: 0.08 second
QT interval: 0.32 to 0.36 second
Interpretation: Sinus rhythm at 65 beats/min with ST-segment elevation (STE)

79. **Fig. 10.9**
Rhythm: Irregular
Rate: About 440 beats/min
P waves: None
PR interval: None
QRS duration: 0.12 to 0.14 second
QT interval: None
Interpretation: PMVT at about 440 beats/min

80. **Fig. 10.10**
Rhythm: Atrial and ventricular rhythms are essentially regular
Rate: Ventricular 45 beats/min; atrial 107 beats/min
P waves: Upright; more Ps than QRSs
PR interval: None
QRS duration: 0.14 to 0.16 second
QT interval: 0.40 to 0.44 second
Interpretation: Third-degree AV block at 45 beats/min with a wide QRS

81. **Fig. 10.11**
Rhythm: Regular
Rate: 83 beats/min
P waves: Inverted before each QRS; artifact is present
PR interval: 0.12 second
QRS duration: 0.08 second
QT interval: 0.36 second
Interpretation: Accelerated junctional rhythm at 83 beats/min with deeply inverted T waves

82. **Fig. 10.12**
Rhythm: Irregular
Rate: About 110 beats/min; 54 beats/min (sinus beats)
P waves: Sinus P waves; none with ventricular beats
PR interval: 0.12 second (sinus beats)
QRS duration: 0.06 to 0.08 second (sinus beats)
QT interval: 0.28 second (sinus beats)
Interpretation: Sinus bradycardia at 54 beats/min with ventricular bigeminy; rate about 110 beats/min if ventricular beats counted in the rate

83. **Fig. 10.13**
Rhythm: Regular
Rate: 150 beats/min
P waves: Flutter waves visible
PR interval: None
QRS duration: 0.08 second
QT interval: Unable to determine
Interpretation: Atrial flutter with a rapid ventricular response of 150 beats/min

84. **Fig. 10.14**
Rhythm: Regular
Rate: 180 beats/min
P waves: None visible
PR interval: Unable to determine
QRS duration: 0.06 second
QT interval: 0.20 to 0.24 second
Interpretation: AVNRT at 180 beats/min

85. **Fig. 10.15**
Rhythm: Regular
Rate: Ventricular 42 beats/min; atrial 71 beats/min
P waves: Upright; more Ps than QRSs
PR interval: None
QRS duration: 0.12 second
QT interval: 0.48 second
Interpretation: Third-degree AV block at 42 beats/min with a wide QRS, a prolonged QT interval, and STE

86. **Fig. 10.16**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Ventricular paced rhythm with 100% capture at 65 pulses/min

87. **Fig. 10.17**
Rhythm: Irregular
Rate: 100 beats/min
P waves: Upright before each QRS; the P wave of beat 7 is early and distorts the T wave of the preceding beat
PR interval: 0.16 second (sinus beats)
QRS duration: 0.04 to 0.06 second (sinus beats)
QT interval: 0.28 to 0.30 second (sinus beats)
Interpretation: Sinus rhythm at 100 beats/min with a PAC; beat 7 is the PAC

88. **Fig. 10.18**
Rhythm: Irregular
Rate: 140 beats/min
P waves: None consistently visible
PR interval: None
QRS duration: 0.06 to 0.10 second
QT interval: 0.28 to 0.32 second
Interpretation: AFib with a ventricular response of 140 beats/min; ST-segment depression in lead II

89. **Fig. 10.19**
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Dual-chamber paced rhythm with 100% capture at 71 pulses/min

90. **Fig. 10.20**
Rhythm: Irregular
Rate: 60 beats/min
P waves: Upright before each QRS; a PQRST complex is missing
PR interval: 0.16 second
QRS duration: 0.06 to 0.08 second
QT interval: 0.32 to 0.36 second
Interpretation: Sinus rhythm at 60 beats/min with an episode of SA block

91. **Fig. 10.21**
Rhythm: Irregular
Rate: 110 beats/min
P waves: Upright before most QRSs; inverted with beats 8 and 10
PR interval: 0.12 to 0.16 second (sinus beats)
QRS duration: 0.06 to 0.08 second (sinus beats)
QT interval: 0.24 to 0.28 second (sinus beats)
Interpretation: Sinus tachycardia at 110 beats/min with two PJCs

92. **Fig. 10.22**
Rhythm: Regular
Rate: 69 beats/min
P waves: Upright; 1:1 relationship
PR interval: 0.36 second
QRS duration: 0.10 to 0.11 second
QT interval: 0.40 to 0.44 second
Interpretation: Sinus rhythm at 69 beats/min with first-degree AV block and STE

93. **Fig. 10.23**
Rhythm: Ventricular irregular; atrial regular
Rate: Ventricular 60 beats/min; atrial 100 beats/min
P waves: Upright; more Ps than QRSs
PR interval: Lengthening
QRS duration: 0.04 second
QT interval: 0.28 second
Interpretation: Second-degree AV block type I at 60 beats/min with STE; although the complexes on the right side of the rhythm strip show 2:1 conduction, comparison of the PR intervals of beats 1 and 2 enable a diagnosis of second-degree type I AV block

94. **Fig. 10.24**
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Dual-chamber paced rhythm with 100% capture at 70 pulses/min

95. **Fig. 10.25**
Rhythm: Regular
Rate: 73 beats/min
P waves: None visible
PR interval: None
QRS duration: 0.12 to 0.14 second
QT interval: 0.32 second
Interpretation: Accelerated idioventricular rhythm at 73 beats/min

96. **Fig. 10.26**
Rhythm: Irregular
Rate: 90 beats/min
P waves: No identifiable P waves; fibrillatory waves present
PR interval: None
QRS duration: 0.06 to 0.08 second
QT interval: Unable to determine; no consistently identifiable T waves
Interpretation: AFib at 90 beats/min

97. **Fig. 10.27**
Rhythm: Irregular
Rate: 60 beats/min
P waves: Upright before most QRSs; none visible with beat 4
PR interval: 0.16 to 0.18 second
QRS duration: 0.08 second
QT interval: 0.44 second
Interpretation: Sinus rhythm at 60 beats/min with an episode of sinus arrest and a junctional escape beat

98. **Fig. 10.28**
Rhythm: Irregular
Rate: 60 beats/min
P waves: Upright before each QRS
PR interval: 0.20 to 0.22 second
QRS duration: 0.06 to 0.08 second
QT interval: 0.36 to 0.38 second
Interpretation: Sinus arrhythmia with first-degree AV block at 60 beats/min; artifact is present

99. **Fig. 10.29**
Rhythm: Irregular
Rate: 80 beats/min
P waves: Upright before each QRS; some are smooth and rounded, others are pointed
PR interval: 0.16 to 0.20 second
QRS duration: 0.08 second
QT interval: 0.36 second
Interpretation: Sinus rhythm at 80 beats/min with PACs (beats 3 and 5 are PACs)

100. **Fig. 10.30**
Rhythm: Irregular
Rate: 120 beats/min
P waves: Upright with sinus beats; none with ventricular beats
PR interval: 0.18 second (sinus beats)

QRS duration: 0.06 second (sinus beats)
QT interval: 0.28 second (sinus beats)
Interpretation: Sinus tachycardia at 120 beats/min with two uniform PVCs

101. Fig. 10.31
Rhythm: Regular
Rate: 130 beats/min
P waves: Upright; 1:1 relationship
PR interval: 0.16 second
QRS duration: 0.06 to 0.08 second
QT interval: 0.28 second
Interpretation: Sinus tachycardia at 130 beats/min with ST-segment depression; artifact is present

102. Fig. 10.32
Rhythm: Regular
Rate: 55 beats/min
P waves: Flutter waves present
PR interval: None
QRS duration: 0.08 second
QT interval: Unable to determine
Interpretation: Atrial flutter at 55 beats/min

103. Fig. 10.33
Rhythm: Regular
Rate: 69 beats/min
P waves: Upright and peaked; 1:1 relationship
PR interval: 0.16 second
QRS duration: 0.08 second
QT interval: 0.32 second
Interpretation: Sinus rhythm at 69 beats/min with ST-segment depression and inverted T waves

104. Fig. 10.34
Rhythm: Regular
Rate: 225 beats/min
P waves: None visible
PR interval: None
QRS duration: 0.06 second
QT interval: Unable to determine
Interpretation: AVNRT at 225 beats/min with ST-segment depression

105. Fig. 10.35
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Ventricular paced rhythm with a PVC, a paced beat, nonsustained monomorphic VT, and a paced beat; paced interval 71 pulses/min

106. Fig. 10.36
Rhythm: Irregular
Rate: 70 beats/min
P waves: Vary in size and shape
PR interval: Varies
QRS duration: 0.08 to 0.12 second
QT interval: 0.40 to 0.44 second
Interpretation: Underlying rhythm is sinus but pacemaker site varies; ventricular rate 70 beats/min; patient with known WPW pattern; note the delta waves

107. Fig. 10.37
Rhythm: Irregular
Rate: 50 beats/min
P waves: Upright; none visible before the QRS with beat 3, but an inverted P wave appears after it
PR interval: 0.24 second
QRS duration: 0.08 second
QT interval: 0.32 to 0.36 second
Interpretation: Sinus rhythm at 50 beats/min with a first-degree AV block, an episode of sinus arrest, a junctional escape beat, and ST-segment depression

108. Fig. 10.38
Rhythm: Ventricular absent; atrial regular
Rate: Ventricular none; atrial 40 beats/min
P waves: Upright
PR interval: None
QRS duration: None
QT interval: None
Interpretation: P-wave asystole

109. Fig. 10.39
Rhythm: None
Rate: None
P waves: None
PR interval: None
QRS duration: None
QT interval: None
Interpretation: Ventricular fibrillation

110. Fig. 10.40
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? Yes—failure to capture and failure to sense
Interpretation: Dual-chamber pacemaker with failure to capture (seventh spike) and failure to sense (eighth spike) at 71 pulses/min

111. Fig. 10.41
Rhythm: Ventricular essentially regular; atrial essentially regular
Rate: Ventricular 37 beats/min; atrial 100 beats/min
P waves: Upright; more Ps than QRSs
PR interval: None
QRS duration: 0.08 to 0.10 second
QT interval: Unable to determine because T waves are not clearly visible
Interpretation: Third-degree AV block at 37 beats/min

112. Fig. 10.42
Rhythm: Regular
Rate: 39 beats/min
P waves: None visible

PR interval: None
QRS duration: 0.18 second
QT interval: 0.32 second
Interpretation: Idioventricular rhythm at 39 beats/min

113. Fig. 10.43
Rhythm: Ventricular regular; atrial regular
Rate: Ventricular 36 beats/min; atrial 108 beats/min
P waves: Upright; more Ps than QRSs; 3:1 relationship
PR interval: 0.16 second
QRS duration: 0.08 to 0.10 second
QT interval: Unable to determine
Interpretation: Advanced second-degree AV block with 3:1 conduction at 36 beats/min

114. Fig. 10.44
Rhythm: Irregular
Rate: 120 beats/min
P waves: Vary in size, shape, and direction
PR interval: Varies
QRS duration: 0.08 to 0.10 second
QT interval: 0.24 to 0.28 second
Interpretation: Multifocal atrial tachycardia at 120 beats/min

115. Fig. 10.45
Rhythm: Irregular
Rate: 50 beats/min
P waves: Upright; the P wave of beat 3 is early
PR interval: 0.16 second
QRS duration: 0.08 second
QT interval: 0.40 to 0.44 second
Interpretation: Sinus rhythm at 50 beats/min with a PAC

116. Fig. 10.46
Rhythm: Irregular
Rate: 100 beats/min (sinus beats); 136 beats/min (ventricular beats)
P waves: Low amplitude but upright with sinus beats
PR interval: 0.16 to 0.20 second (sinus beats)
QRS duration: 0.08 second (sinus beats); 0.12 second (ventricular beats)
QT interval: 0.34 second (sinus beats)
Interpretation: Sinus rhythm at 100 beats/min with a run of monomorphic VT at 136 beats/min

117. Fig. 10.47
Rhythm: Irregular
Rate: 68 beats/min (sinus beats); 75 beats/min (junctional beats)
P waves: Upright with sinus beats; inverted with junctional beats
PR interval: 0.16 second (sinus beats)
QRS duration: 0.08 second
QT interval: 0.28 to 0.32 second
Interpretation: Sinus rhythm at 68 beats/min to an accelerated junctional rhythm at 75 beats/min

118. Fig. 10.48
Rhythm: Irregular
Rate: 50 beats/min (top strip)
P waves: Vary in size, shape, and direction
PR interval: Varies
QRS duration: 0.06 to 0.08 second
QT interval: 0.40 to 0.44 second
Interpretation: Wandering atrial pacemaker at 50 beats/min

119. Fig. 10.49
Rhythm: Ventricular irregular; atrial regular
Rate: Ventricular 50 beats/min; atrial 167 beats/min
P waves: Upright and notched; more Ps than QRSs
PR interval: 0.26 second; the PR intervals before and after the nonconducted P waves are constant
QRS duration: 0.12 to 0.14 second
QT interval: Unable to determine
Interpretation: Second-degree AV block type II at 60 beats/min with ST-segment depression

120. Fig. 10.50
Rhythm: Regular
Rate: 150 beats/min
P waves: One P wave appears after each QRS
PR interval: None
QRS duration: 0.06 second
QT interval: 0.20 second
Interpretation: Junctional tachycardia at 150 beats/min

121. Fig. 10.51
Rhythm: Irregular
Rate: 83 beats/min (sinus beats)
P waves: Flutter waves initially present; sinus P waves present for last 3 beats
PR interval: 0.20 second (sinus beats)
QRS duration: 0.08 to 0.10 second (sinus beats)
QT interval: 0.36 to 0.40 second (sinus beats)
Interpretation: Atrial flutter with a period of asystole to sinus rhythm at 83 beats/min

122. Fig. 10.52
Rhythm: Regular
Rate: 45 beats/min
P waves: Low amplitude but upright
PR interval: 0.20 to 0.24 second
QRS duration: 0.10 to 0.12 second
QT interval: 0.36 to 0.40 second
Interpretation: Sinus bradycardia at 45 beats/min with first-degree AV block and horizontal ST segments; artifact is present

123. Fig. 10.53
Rhythm: Irregular
Rate: 110 beats/min
P waves: Upright; the P wave of beat 5 is early and inverted
PR interval: 0.16 second (sinus beats)
QRS duration: 0.06 second (sinus beats)
QT interval: Unable to determine
Interpretation: Sinus tachycardia at 110 beats/min with a PJC (beat 5 is the PJC)

124. Fig. 10.54
Rhythm: Ventricular regular; atrial regular
Rate: Ventricular 40 beats/min; atrial 79 beats/min
P waves: Upright; more Ps than QRSs
PR interval: 0.28 second
QRS duration: 0.06 to 0.08 second
QT interval: 0.48 second (prolonged)
Interpretation: 2:1 AV block at 40 beats/min with a prolonged QT interval

125. Fig. 10.55
Rhythm: Irregular
Rate: 30 beats/min (junctional beats)
P waves: Upright (sinus beats); none with junctional beats
PR interval: 0.16 second (sinus beats); none (junctional beats)
QRS duration: 0.06 to 0.08 second
QT interval: 0.36 to 0.38 second
Interpretation: Sinus beat, two junctional beats, sinus beat; ST-segment depression; artifact is present

126. Fig. 10.56
Rhythm: Irregular
Rate: 80 beats/min
P waves: Upright; the P wave of beat 3 is early and distorts the previous T wave
PR interval: 0.16 second
QRS duration: 0.06 second
QT interval: Unable to determine
Interpretation: Sinus rhythm at 80 beats/min with a PAC and STE

127. Fig. 10.57
Rhythm: Regular
Rate: 79 beats/min
P waves: None visible
PR interval: None
QRS duration: 0.06 second
QT interval: 0.36 to 0.38 second
Interpretation: Accelerated junctional rhythm at 79 beats/min

128. Fig. 10.58
Rhythm: Irregular
Rate: 40 beats/min
P waves: Upright; some are notched
PR interval: 0.20 second
QRS duration: 0.08 second
QT interval: 0.44 second
Interpretation: Sinus bradycardia at 40 beats/min with an episode of sinus arrest

129. Fig. 10.59
Rhythm: Irregular
Rate: 90 beats/min
P waves: Fibrillatory waves are present
PR interval: None
QRS duration: 0.08 to 0.10 second
QT interval: Unable to determine
Interpretation: AFib at 90 beats/min

130. Fig. 10.60
Rhythm: Irregular
Rate: 56 beats/min (sinus beats)
P waves: Upright with sinus beats; none with ventricular beats
PR interval: 0.16 second (sinus beats)
QRS duration: 0.06 second (sinus beats)
QT interval: Unable to determine
Interpretation: Sinus bradycardia at 56 beats/min with multiform ventricular bigeminy, a run of VT, and ST-segment depression

131. Fig. 10.61
Rhythm: Irregular
Rate: 130 beats/min
P waves: Vary in size, shape, and direction
PR interval: Varies
QRS duration: 0.08 second
QT interval: Varies
Interpretation: Multifocal atrial tachycardia at 130 beats/min

132. Fig. 10.62
Rhythm: Irregular
Rate: 50 beats/min
P waves: Upright; more Ps than QRSs
PR interval: Lengthens (visible in last three beats)
QRS duration: 0.08 to 0.10 second
QT interval: 0.40 to 0.44 second
Interpretation: Second-degree AV block type I at 50 beats/min with ST-segment depression; note the presence of 2:1 AV block at the start of the rhythm strip

133. Fig. 10.63
Rhythm: Ventricular regular; atrial regular
Rate: Ventricular 34 beats/min; atrial 68 beats/min
P waves: Upright; more Ps than QRSs; 2:1 relationship
PR interval: 0.14 to 0.16 second
QRS duration: 0.10 second
QT interval: 0.52 second (prolonged)
Interpretation: 2:1 AV block at 34 beats/min with ST-segment depression, a prolonged QT interval, and tall T waves

134. Fig. 10.64
Rhythm: Irregular
Rate: 60 beats/min
P waves: Upright; early P wave after beat 3; no P wave with beat 4
PR interval: 0.22 to 0.24 second
QRS duration: 0.08 second (sinus beats)
QT interval: 0.36 to 0.40 second
Interpretation: Sinus rhythm at 60 beats/min with a first-degree AV block, a nonconducted PAC, and a ventricular escape beat

135. Fig. 10.65
Rhythm: Irregular
Rate: 50 beats/min
P waves: Upright before each QRS; a PQRST complex is missing
PR interval: 0.16 second
QRS duration: 0.08 second
QT interval: 0.32 to 0.36 second
Interpretation: Sinus bradycardia at 50 beats/min with an episode of SA block

136. Fig. 10.66
Rhythm: Irregular
Rate: About 330 beats/min
P waves: None
PR interval: None
QRS duration: Varies
QT interval: Unable to determine
Interpretation: PMVT at about 330 beats/min

137. Fig. 10.67
Rhythm: Regular
Rate: Ventricular 90 beats/min; atrial 180 beats/min
P waves: Upright before each QRS; others are hidden in the T waves; 2:1 relationship
PR interval: 0.16 second
QRS duration: 0.06 second
QT interval: 0.24 to 0.28 second
Interpretation: Atrial tachycardia at 90 beats/min with 2:1 block and STE

138. Fig. 10.68
Rhythm: Regular
Rate: 88 beats/min
P waves: Upright
PR interval: 0.16 to 0.20 second
QRS duration: 0.10 to 0.12 second (notched)
QT interval: 0.32 second
Interpretation: Sinus rhythm at 88 beats/min with an incomplete BBB

139. Fig. 10.69
Rhythm: Regular
Rate: 60 beats/min
P waves: None visible
PR interval: None
QRS duration: 0.12 to 0.14 second
QT interval: 0.28 to 0.32 second
Interpretation: Accelerated idioventricular rhythm at 60 beats/min with tall T waves; artifact is present

140. Fig. 10.70
Rhythm: Ventricular irregular; atrial regular
Rate: Ventricular 30 beats/min; atrial 79 beats/min
P waves: Upright; more Ps than QRSs
PR interval: 0.14 second
QRS duration: 0.16 second
QT interval: 0.40 to 0.42 second
Interpretation: Advanced second-degree AV block at 30 beats/min with a wide QRS and STE

141. Fig. 10.71
Rhythm: Regular
Rate: 115 beats/min
P waves: Upright; 1:1 relationship
PR interval: 0.16 second
QRS duration: 0.08 second
QT interval: 0.28 second
Interpretation: Sinus tachycardia at 115 beats/min with STE in V_4; artifact is present

142. Fig. 10.72
Rhythm: Irregular
Rate: 60 beats/min
P waves: Upright (sinus beats); inverted after the QRS in beat 5
PR interval: 0.14 second (sinus beats)
QRS duration: 0.06 second
QT interval: 0.40 to 0.44 second
Interpretation: Sinus rhythm at 60 beats/min with a PJC

143. Fig. 10.73
Rhythm: Regular
Rate: 45 beats/min
P waves: None visible
PR interval: None
QRS duration: 0.08 to 0.11 second
QT interval: 0.48 to 0.52 second (prolonged)
Interpretation: Junctional rhythm at 45 beats/min with STE and a prolonged QT interval

144. Fig. 10.74
Rhythm: Irregular
Rate: 40 beats/min
P waves: Upright (sinus beats); none with beat 2
PR interval: 0.36 second (sinus beats)
QRS duration: 0.06 second (sinus beats)
QT interval: 0.36 to 0.38 second
Interpretation: Sinus bradycardia at 40 beats/min with first-degree AV block and a PJC

145. Fig. 10.75
Rhythm: Irregular
Rate: 71 beats/min (sinus beats)
P waves: Upright before each QRS; one early P wave distorts the T wave of beat 4 (most clearly seen in lead MCL_1)
PR interval: 0.20 second (lead II)
QRS duration: 0.12 second (lead II)
QT interval: 0.40 to 0.44 second (lead II)
Interpretation: Sinus rhythm at 71 beats/min with a wide QRS and a nonconducted PAC